

SingleRAN

Bandwidth Sharing of Multimode Base Station Co-Transmission Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 03 (2025-09-26).

Technical Changes

None

Editorial Changes

Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and New Radio (NR).

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
GS M	MRFD-121115	Bandwidth sharing of MBTS Multi-mode Co-Transmission(GBTS)	4 Bandwidth Sharing of Multimode Base Station Co-Transmission
UM TS	MRFD-121125	Bandwidth sharing of MBTS Multi-mode Co-Transmission(NodeB)	
LTE FDD	MRFD-121135	Bandwidth sharing of MBTS Multi-mode Co-Transmission(eNodeB)	
LTE TDD	MRFD-121145	Bandwidth sharing of MBTS Multi-mode Co-Transmission (LTE TDD)	
NB- IoT	MRFD-121155	Bandwidth sharing of MBTS Multi-mode Co-Transmission(NB-IoT)	
NR	MRFD-151168	Bandwidth sharing of MBTS Multi-mode Co-Transmission(NR)	

 NOTE

In this document, LTE, eNodeB, and eRAN always include FDD, TDD, and NB-IoT. Otherwise, LTE FDD, LTE TDD, LTE NB-IoT, LTE FDD eNodeB, LTE TDD eNodeB, LTE NB-IoT eNodeB, LTE FDD eRAN, LTE TDD eRAN, and LTE NB-IoT eRAN will be used. The "L", "T", and "M" in RAT acronyms refer to LTE FDD, LTE TDD, and LTE NB-IoT, respectively.

In this document, NR and gNodeB always include FDD and TDD.

3 Introduction

Background

For a separate-MPT multimode base station in co-transmission scenarios, the co-transmission port transmits and receives the local data and the passing data. In this case, the co-transmission port centrally schedules and manages the data of multiple RATs. For a co-MPT multimode base station in co-transmission scenarios, the co-transmission port transmits and receives only the local data, which includes the data for all RATs of this base station. In this case, the co-transmission port centrally schedules and manages the data for all modes.

To enable a co-transmission port to implement unified data scheduling and management, differentiation and fairness among different service types and RATs must be ensured. Moreover, transmission resource congestion when all of the RATs have overlapping traffic bursts must be addressed. Therefore, Huawei introduces the Bandwidth Sharing of Multimode Base Station Co-Transmission feature. The differentiation and fairness are described as follows:

- **Differentiation:** Transmission differentiation is used when transmission bandwidth is limited. Transmission differentiation prioritizes bandwidth use, with real-time services taking precedence over non-real-time services.
- **Fairness:** If transmission congestion occurs, service differentiation ensures that real-time services are preferentially processed. As a result, non-real-time services may experience packet losses, which affect fairness among non-real-time services. The transmission flow control function enables each type of service or each RAT to be allocated a certain amount of bandwidth. This eliminates the possibility that a certain service or a certain RAT experiences service interruptions because of lack of transmission bandwidth.

This feature adopts four recommended transmission resource management strategies: mapping between traffic classes and transmission priorities, traffic limiting and shaping, load control, and flow control. For details about transmission resource management strategies for GSM, UMTS, LTE, and NR, see *Transmission Resource Management* for GBSS, WCDMA RAN, eRAN, and 5G RAN.

Transmission Configuration Model

As network traffic volume increases, network transmission scenarios become increasingly complicated. For example, transmission ports or boards need to be adjusted to support inter-board functions. The traditional transmission network

configuration is complicated and struggles to meet the requirement for fast deployment on the live network. Therefore, a new transmission configuration model is introduced to decouple transmission configurations from physical devices to simplify operations.

A transmission configuration model can either be new or old, which is controlled by the **GTRANSPARA.TRANS CFGMODE** parameter.

- When this parameter is set to **OLD**, the old transmission configuration model (referred to as old model) is used. In the old model, location information such as cabinet, subrack, and slot numbers are configured, the transmission configuration is bound to physical devices, and IPv4 and IPv6 configuration objects above the IP layer are independent of each other.
- When this parameter is set to **NEW**, the new transmission configuration model (referred to as new model) is used. In the new model, the transmission model is decoupled from the device model. That is, the transmission configuration does not contain cabinet, subrack, and slot numbers, and IPv4 and IPv6 configuration objects above the IP layer are combined. This model facilitates the expansion of new transmission functions and requires fewer transmission configuration parameters.

Compared with the old model, the new model incorporates an **INTERFACE** MO to isolate the upper layer from the physical layer. In this way, transmission configuration objects are decoupled from physical devices.

In addition, with the new model, when a transmission link is configured or a transmission board or port is modified, the information such as the cabinet, subrack, and slot numbers as well as the subboard type only needs to be configured for the physical layer and data link layer.

The new model is applicable to base stations excluding GBTs and pico base stations.

4 Bandwidth Sharing of Multimode Base Station Co-Transmission

4.1 Principles

4.1.1 Introduction

The Bandwidth Sharing of Multimode Base Station Co-Transmission feature centrally manages GSM, UMTS, LTE, and NR transmission resources. When transmission resources are congested, this feature ensures the smooth processing of high-priority services and prevents GSM, UMTS, LTE, and NR services from impacting each other. This ensures high service quality and good user experience. This feature performs transmission resource management through the following strategies: mapping between traffic classes and transmission priorities, traffic limiting and shaping, load control, and flow control.

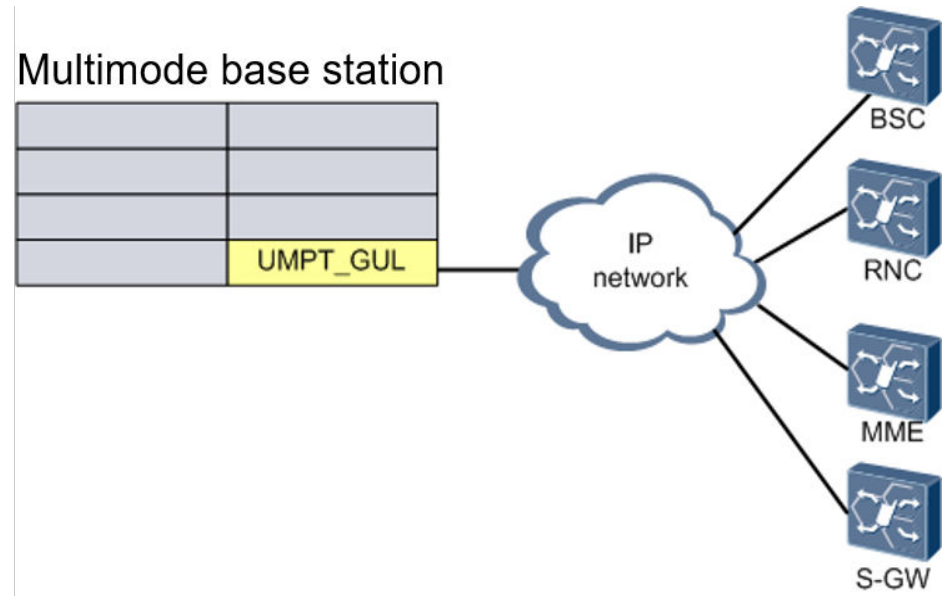
If this feature is not enabled, the transmission resources for each RAT of a multimode base station are managed in the same way as those of a single-mode base station. For details about transmission resource management strategies for single-mode base stations, see *Transmission Resource Management* for GBSS, WCDMA RAN, eRAN, and 5G RAN.

4.1.2 Application Networking

This feature applies to networking schemes where both the local end (the multimode base station) and the peer end (the base station controller, MME, or S-GW) use IP transmission (IP over FE/GE or IP over E1/T1).

Figure 4-1 shows the networking scheme for a GUL co-MPT base station in co-transmission scenarios.

Figure 4-1 Networking scheme for a GUL co-MPT base station in co-transmission scenarios



For details about the networking scheme for a multimode base station in co-transmission scenarios, see *Common Transmission* for SingleRAN.

NOTE

- In this document, a multimode base station can be a GU, GL, GT, GN (5G SA networking), UL, UT, UN (5G SA networking), LT, LN (5G SA/NSA networking), or TN (5G SA/NSA networking) dual-mode base station, or a GUL, GUT, GUN (5G SA networking), GLT, GLN (5G SA/NSA networking), GTN (5G SA/NSA networking), ULT (5G SA/NSA networking), ULN (5G SA/NSA networking), UTN (5G SA/NSA networking), LTN (5G SA/NSA networking), GULT (5G SA/NSA networking), GULN (5G SA/NSA networking), GUTN (5G SA/NSA networking), GLTN (5G SA/NSA networking), ULTN (5G SA/NSA networking), or GULTN (5G SA/NSA networking) multimode base station. The GSM, UMTS, and LTE sides of a multimode base station are referred to as the GBTS/eGBTS, NodeB, and eNodeB, respectively. LTE can be LTE FDD, LTE TDD, or LTE NB-IoT. The NR side of a multimode base station is referred to as the gNodeB.
- Multimode base stations are classified into co-MPT and separate-MPT multimode base stations. The GSM side of a separate-MPT multimode base station can be either an eGBTS or a GBTS. The GSM side of a co-MPT multimode base station must be an eGBTS. The GBTS is not recommended for providing a co-transmission port to a separate-MPT multimode base station. This scenario is not covered in this document.

4.1.3 Transmission Strategies

4.1.3.1 Transmission Priorities

In the Bandwidth Sharing of Multimode Base Station Co-Transmission feature, the mapping between GSM/UMTS/LTE/NR traffic classes and transmission priorities must be configured to ensure that differentiated services (DiffServ) are provided based on transmission priorities. Transmission priorities include the DiffServ Code Point (DSCP), virtual local area network (VLAN) priority, and queue priority.

For NB-IoT, if control plane Clot EPS optimization is used between NB-IoT base stations and the core network, NB-IoT services are carried on SCTP links. The

transmission priority mapping for NB-IoT services is the transmission priority mapping for SCTP services.

DSCP

DSCP is a field in an IP packet header to indicate the QoS requirements. The L3 network implements DiffServ based on the DSCP value.

A multimode base station or a multimode base station controller sets the DSCP value for each IP packet based on the QoS requirements of each service type. The L3 transmission device differentiates each type of service and its QoS requirements according to the DSCP value, and then provides services such as resource allocation, queue scheduling, and packet discarding policies. This service is called per-hop behavior (PHB). All the network nodes for DiffServ comply with the PHB according to the DSCP field contained in an IP packet.

The following table describes how to use MML commands to configure the mapping between traffic classes and DSCP values for each type of base station.

NE	Command	Description
GBTS	SET BTSVLAN	Used to set the mapping from the management-, control-, and user-plane data to DSCP values.
eGBTS and NodeB	SET DIFPRI	Used to set the mapping from management- and control-plane data to DSCP values.
	ADD SCTPLNK and ADD SCTPTEMPLATE	Used to set the mapping from control-plane data to DSCP values.
	ADD TRMMAP and SET PHBMAP	Used to set the mapping from user-plane data to DSCP values for the eGBTS and NodeB on the BSC/RNC side.
eNodeB and gNodeB	SET DIFPRI	Used to set the mapping from management- and control-plane data to DSCP values.
	ADD SCTPLNK and ADD SCTPTEMPLATE	Used to set the mapping from control-plane data to DSCP values.
	MOD UDTPARAGRP	Used to set the mapping from user-plane data to DSCP values.
NOTE The ADD SCTPLNK command is used in link configuration mode, and the ADD SCTPTEMPLATE command is used in endpoint configuration mode. The eGBTS supports only the link configuration mode. The gNodeB supports only the endpoint configuration mode. The NodeB and eNodeB support both the link configuration mode and endpoint configuration mode.		

Pay attention to the following when configuring the mapping from the management-, control-, and user-plane data to DSCP values.

- For separate-MPT multimode base stations in co-transmission scenarios, run the corresponding commands to configure the mapping from the management- and control-plane data to DSCP values for the GBTS, eGBTS, NodeB, eNodeB, and gNodeB. For co-MPT multimode base stations in co-transmission scenarios, run the **SET DIFPRI** command to configure the mapping from the management- and control-plane data to DSCP values for the eGBTS, NodeB, and eNodeB. For co-transmission on a co-MPT multimode base station, you can run the **ADD SCTPLNK/MOD SCTPLNK** command (in link configuration mode) or the **ADD SCTPTemplate/MOD SCTPTemplate** command (in endpoint configuration mode) to separately configure the RAT-specific mapping between signaling and DSCP values.
- For multimode base stations in co-transmission scenarios, run the corresponding commands to configure the RAT-specific mapping from the user-plane data to DSCP values for the GBTS, eGBTS, NodeB, eNodeB, and gNodeB.
- The mapping between traffic classes and DSCP values for NR, LTE, UMTS, and GSM should be consistent on the base station, the base station controller, and the core network. For details, see *Transmission Resource Management* for GSM BSS, WCDMA RAN, eRAN, and 5G RAN.
- If control plane Clot EPS optimization is used between NB-IoT base stations and the core network, NB-IoT services are carried only on SCTP links. The transmission priority mapping for NB-IoT services is the transmission priority mapping for SCTP services.

VLAN Priority

The VLAN tag defines the VLAN priority of an IP packet. Based on the VLAN priority, Layer 2 devices can implement DiffServ.

VLAN priorities of packets with different traffic classes are determined by DSCP values. [Table 4-1](#) provides the default mapping between DSCP values and VLAN priorities on the multimode base station side.

Table 4-1 Default mapping between DSCP values and VLAN priorities

DSCP	VLAN Priority
0-7	0
8-15	1
16-23	2
24-31	3
32-39	4
40-47	5
48-55	6
56-63	7

Queue Priority

Queue priority defines the scheduling priority of a queue. Each Ethernet port or PPP link has eight queues. The default scheduling sequence is as follows: PQ (Priority Queuing) 1 > PQ2 > PQ3 > WRR (Weighted Round Robin). WRR includes WFQ (Weighted Fair Queuing) 4, WFQ5, WFQ6, WFQ7, and WFQ8. A multimode base station puts packets with different traffic classes into different queues to implement DiffServ.

Queue priorities are determined for different traffic types according to the mapping between DSCP values and queue priorities. [Table 4-2](#) and [Table 4-3](#) list the default mapping between DSCP values of NR, LTE, UMTS, and GSM services and queue priorities for multimode base stations. You are not advised to modify the default mapping.

Table 4-2 Default mapping between DSCP values and queue priorities for the GBTS

DSCP	Queue	Queue Priority
40-63	PQ1	0
Reserved	PQ2	1
Reserved	PQ3	2
32-39	WFQ4	3
24-31	WFQ5	3
16-23	WFQ6	3
8-15	WFQ7	3
0-7	WFQ8	3

Table 4-3 Default mapping between DSCP values and queue priorities for the eGBTS, NodeB, eNodeB, gNodeB, and co-MPT multimode base station

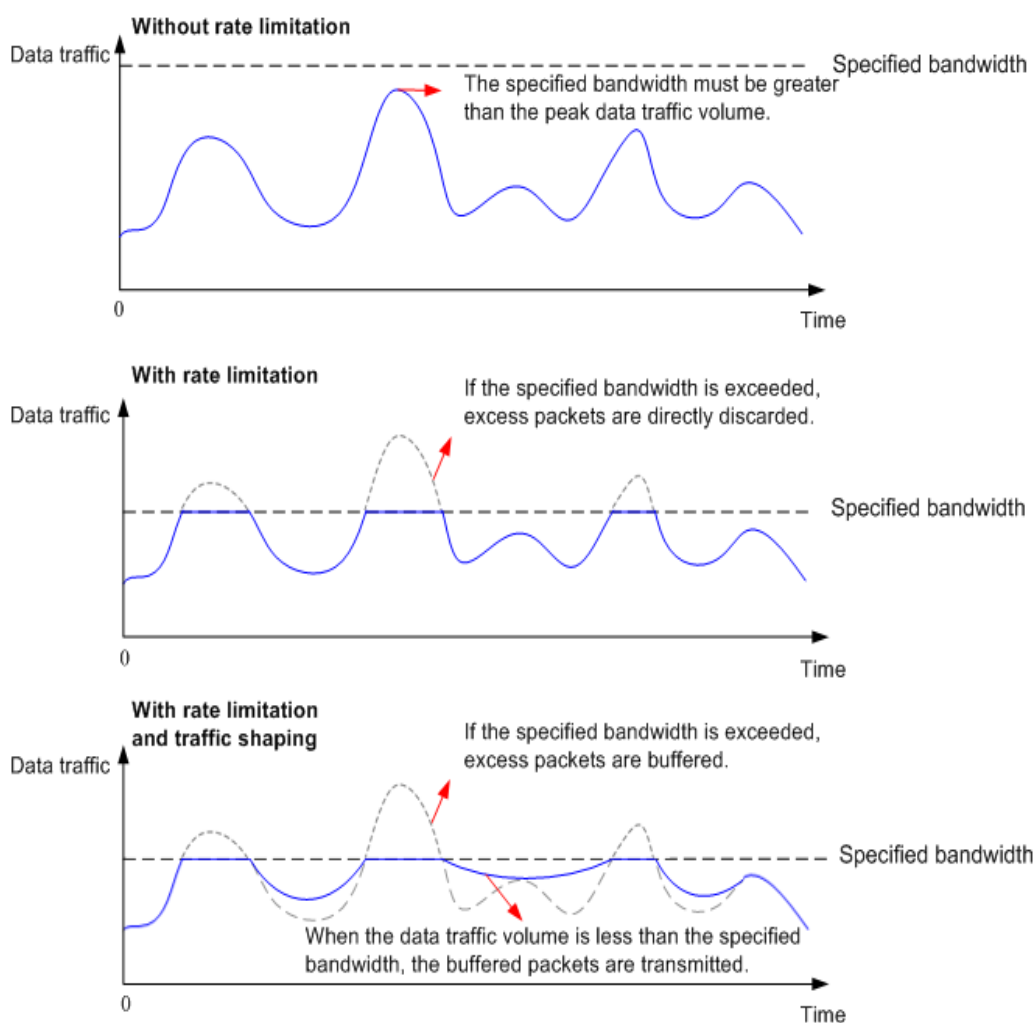
DSCP	Queue	Queue Priority
48-63	PQ1	0
40-47	PQ2	1
32-39	PQ3	2
24-31	WFQ4	3
16-23	WFQ5	3
8-15	WFQ6	3

DSCP	Queue	Queue Priority
0-7	WFQ7	3

4.1.3.2 Traffic Limiting and Shaping

When transmission network resources are limited, transmission devices may be incapable of receiving excess packets that arrive at the co-transmission port in a multimode base station. To prevent transmission devices from discarding packets, the traffic limiting function is introduced. Data services have unstable transmission rates due to unexpected traffic bursts. The traffic shaping function is introduced to ensure stable rates in a multimode base station. The traffic limiting and shaping functions use the Generic Traffic Shaping (GTS) technology, which shapes irregular data flows, or traffic flows that do not match preset characteristics, to balance the bandwidths between upstream and downstream nodes. See [Figure 4-2](#). The traffic limiting and shaping functions can limit the traffic and reduce packet loss caused by traffic bursts.

Figure 4-2 Data traffic without rate limitation, with rate limitation, and with rate limitation and shaping



Base stations cannot dynamically adjust the data rates of real-time services. To prevent real-time service congestion, at the early stage of network deployment, the minimum bandwidth planned for transmission devices must be greater than the total bandwidth planned for real-time services in a GU/GL/UL/GUL/LN/GLN/ULN/GULN multimode base station.

For non-real-time services, if transmission network resources are insufficient, traffic limiting and shaping can be performed on non-real-time services of GSM, UMTS, LTE, and NR to avoid congestion.

Separate-MPT multimode base stations and co-MPT multimode base stations support traffic limiting and shaping at the base station level and logical port level.

Base-station-level traffic limiting and shaping:

If the eGBTS/NodeB/eNodeB/gNodeB/co-MPT multimode base station provides a co-transmission port, you can run the **SET LR** (in the old model)/**ADD PORTLR** (in the new model) command and specify the **LR.CIR** (in the old model)/**PORTLR.CIR** (in the new model) parameter to set the bandwidth after traffic limiting for a base station.

Logical-port-level traffic limiting and shaping:

- For separate-MPT/co-MPT multimode base stations
If the eGBTS/NodeB/eNodeB/gNodeB/co-MPT multimode base station provides a co-transmission port, you can run the **ADD RSCGRP** (in the old model)/**ADD IPRSCGRP** (in the new model) command and specify the **RSCGRP.TXBW** (in the old model)/**IPRSCGRP.TXBW** (in the new model) parameter to set the bandwidth after traffic limiting for a logical port.
- For multimode base station controllers
You can run the **ADD IPLOGICPORT** command and specify the **IPLOGICPORT.CIR** parameter to set the bandwidth after traffic limiting for a logical port.

Pay attention to the following:

- A transmission resource group can be configured as a default port transmission resource group or a non-default port transmission resource group. A physical port can be configured with one default port transmission resource group and multiple non-default port transmission resource groups. When a co-MPT base station is used, the following transmission resource group configuration policies are recommended:
 - All modes use the same default port transmission resource group for rate limitation and traffic shaping.
 - Each mode uses a different non-default port transmission resource group to implement rate limitation and traffic shaping.
- When a separate-MPT base station is used, the following transmission resource group configuration policies are recommended:
 - All modes use the same default port transmission resource group for rate limitation and traffic shaping. (The board providing transmission ports is of high specifications.)
 - Each mode uses a different non-default port transmission resource group to implement rate limitation and traffic shaping. (The board providing transmission ports is of high specifications.)

- When co-transmission is used in secure networking scenarios, the following transmission resource group configuration policies are recommended:
 - All modes use the same default port transmission resource group for rate limitation and traffic shaping in a co-MPT base station.
 - Each mode uses a different non-default port transmission resource group to implement rate limitation and traffic shaping in a co-MPT or separate-MPT base station.
- You are not advised to modify the rate using the **ADD ETHPORT/SET ETHPORT** command.

4.1.3.3 Load Control

Load control consists of the following functions:

- Admission control
Ensures the quality of admitted services by preventing excessive admission.
- Load reshuffling (LDR)
Increases the admission success rate and system capacity by relieving transmission load and preventing transmission resource congestion. LTE does not support LDR.
- Overload control (OLC)
Alleviates the negative impact of overload on high-priority users by quickly reducing transmission load.

Load control for each mode in a multimode base station in co-transmission scenarios is the same as load control in a single-mode base station. GSM and UMTS loads are controlled by the related base station controller and LTE load is controlled by the eNodeB. NB-IoT and NR services do not support load control. For details about load control for GSM, UMTS, and LTE services, see *Transmission Resource Management* in *GBSS Feature Documentation*, *RAN Feature Documentation*, and *eRAN Feature Documentation*, respectively.

4.1.3.4 Flow Control

When the transmission bandwidth changes dynamically, the available bandwidth of the bottleneck node may be less than the limited bandwidth of the shared port. In this case, if the base station continues to send data based on the limited bandwidth, congestion occurs on the transport network. The flow control algorithm estimates the bottleneck bandwidth of the network based on the detection of the transmission quality, and then dynamically adjusts the transmit (TX) bandwidth to ensure that it does not exceed the bottleneck bandwidth. GSM, UMTS, LTE, and NR support the flow control algorithm in different ways:

- The GBTS/eGBTS and GBSC do not support the flow control algorithm.
- NR currently does not support the flow control algorithm.
- NB-IoT services do not support flow control.
- The eNodeB supports the flow control algorithm in IPv4 transmission, and this algorithm is disabled by default. It is not supported in IPv6 transmission.
- The NodeB and RNC support the flow control algorithm, which is also called the dynamic flow control algorithm.

- For an LN/ULN/GULN/GULTN co-MPT multimode base station that shares transmission resource groups, dynamic bandwidth adjustment based on link detection is not supported by LTE and NR.

The dynamic flow control algorithm of the NodeB performs traffic shaping by detecting the transmission delay, packet loss, and available bandwidth to prevent packet loss caused by congestion on the lub interface. The dynamic flow control algorithm applies to HSUPA and HSDPA services but not to R99 services. There are two types of NodeB dynamic flow control algorithms, as described in [Table 4-4](#).

In the event of transmission resource group sharing within a multimode base station, if the congestion control switch or the dynamic TX bandwidth adjustment switch (specified by **RSCGRPALG.TXBWASW** for old models and **IPRSCGRPALG.TXBWASW** for new models) is turned on, the dynamic flow control algorithm uses only the local bandwidth adjusted based on flow control as the input. Services in the transmission resource group fairly preempt resources based on their transmission priorities. The TX bandwidth cannot be dynamically adjusted by running the **SET LR** (old models) or **ADD PORTLR** (new models) command.

Table 4-4 Dynamic flow control algorithms on the NodeB side

Dynamic Flow Control Algorithm	Control Switch	Description
NodeB uplink bandwidth adaptive adjustment algorithm	Congestion control switch: ULFLOWCTRLPARA.TNLCONGCTRLSWITCH (in the old model)/ IPULFLOWCTRLPARA.TNLCONGCTRLSWITCH (in the new model) Backpressure algorithm switch: RSCGRPALG.TCSW (in the old model)/ IPRSCGRPALG.TCSW (in the new model)	For details, see <i>Transmission Resource Management in RAN Feature Documentation</i> .
NodeB HSDPA adaptive flow control algorithm	Flow control switch: DLFLOWCTRLPARA.SWITCH (in the old model)/ IPDLFLOWCTRLPARA.SWITCH (in the new model)	

Precautions for Configuring the NodeB HSDPA Adaptive Flow Control Algorithm (Downlink)

After the HSDPA adaptive flow control algorithm is enabled on the UMTS side, you are advised to enable enhanced HSDPA flow control and enhanced HSUPA lub flow control in the case of lub congestion to prevent the flow control algorithm from falsely decreasing the bandwidth when packet loss occurs on the

transmission network due to non-congestion reasons (for example, packet loss caused by bit errors).

- For details about enhanced HSDPA flow control, see *HSDPA* in *RAN Feature Documentation*.
- For details about the implementation principle and configuration of enhanced HSUPA lub flow control in the case of lub congestion, see *Transmission Resource Management* in *RAN Feature Documentation*.

In UL/UN/UT/GUL/GUT/ULT/UTN/ULN/GULT/GUTN/ULTN/GULN/GUN/GULTN co-transmission scenarios, as shown in [Table 4-5](#), to prevent LTE and NR services from occupying all the bandwidths released by UMTS HSDPA services during transmission flow control, which may lead to a sharp decrease in the UMTS service bandwidth, you are advised to set **DLFLOWCTRLPARA.FAIRSWITCH** (in the old model)/**IPDLFLOWCTRLPARA.FAIRSWITCH** (in the new model) to **ON** on the NodeB side to protect the bandwidth of UMTS HSDPA services.

IPv6 does not support fair flow control.

Table 4-5 Fair flow control switches on the NodeB side in co-transmission scenarios

Co-Transmission Scenario	Whether the Fair Flow Control Switch Takes Effect
UL/UT/GUL/GUT/ULT/UTN/ULN/GULT/GUTN/ULTN/GULN/GUN/GULTN separate-MPT multimode base station using panel interconnection	No
UL/UT/UN/GUL/GUT/ULT/UTN/ULN/GULT/GUTN/ULTN/GULN/GUN/GULTN co-MPT multimode base station	The fair flow control switch DLFLOWCTRLPARA.FAIRSWITCH (in the old model)/IPDLFLOWCTRLPARA.FAIRSWITCH (in the new model) takes effect only if the NodeB configurations meet the following conditions: • Link mode: The value of the PT parameter in the DLFLOWCTRLPARA (in the old model)/IPDLFLOWCTRLPARA (in the new model) MO is the same as that of the PT parameter in the IPPATH MO of UMTS. • Endpoint mode: The value of the PT parameter in the DLFLOWCTRLPARA (in the old model)/IPDLFLOWCTRLPARA (in the new model) MO is the same as that of the PT parameter of the port corresponding to the local IP address configured in the USERPLANEHOST MO of UMTS.
UL/UT/UN/GUL/GUT/ULT/UTN/ULN/GULT/GUTN/ULTN/GULN/GUN/GULTN separate-MPT multimode base station using backplane interconnection	

It is recommended that the **DLFLOWCTRLPARA.FAIRRATIO** (in the old model)/**IPDLFLOWCTRLPARA.FAIRRATIO** (in the new model) parameter be set to a value between 30% (included) and 70% (included). Otherwise, the actual bandwidth of UMTS HSDPA services may be inconsistent with the guard bandwidth configured for fair flow control. The default value of the **DLFLOWCTRLPARA.FAIRRATIO** (in the old model)/**IPDLFLOWCTRLPARA.FAIRRATIO** (in the new model) parameter is equal to 30% of the actual receive bandwidth of the base station. That is, when the total bandwidth of UMTS HSDPA services decreases to 30% of the actual receive bandwidth of the base station, rate reduction will no longer be performed on these services.

The fair flow control switch can be configured on either a physical port or a loopback port (also called a logical port). It is recommended that the fair flow control switch be configured on a physical port. When configured on the logical port, the fair flow control switch for co-MPT multimode base stations applies only to the following scenarios:

- Scenario 1: One loopback port corresponds to one physical port, and UMTS, LTE, and NR services are carried on the same physical port, as shown in [Figure 4-3](#).
- Scenario 2: One loopback port corresponds to multiple physical ports, and LTE, UMTS, and NR services are carried on different physical ports, as shown in [Figure 4-4](#).

Figure 4-3 Scenario 1

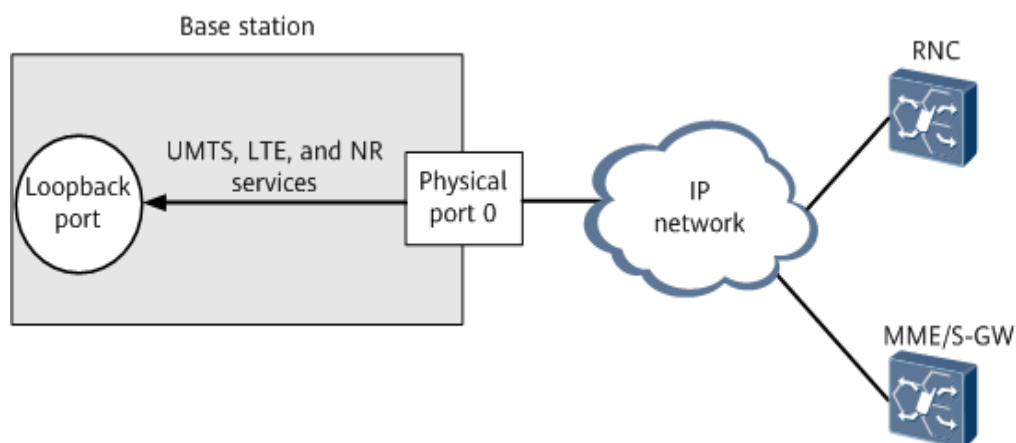
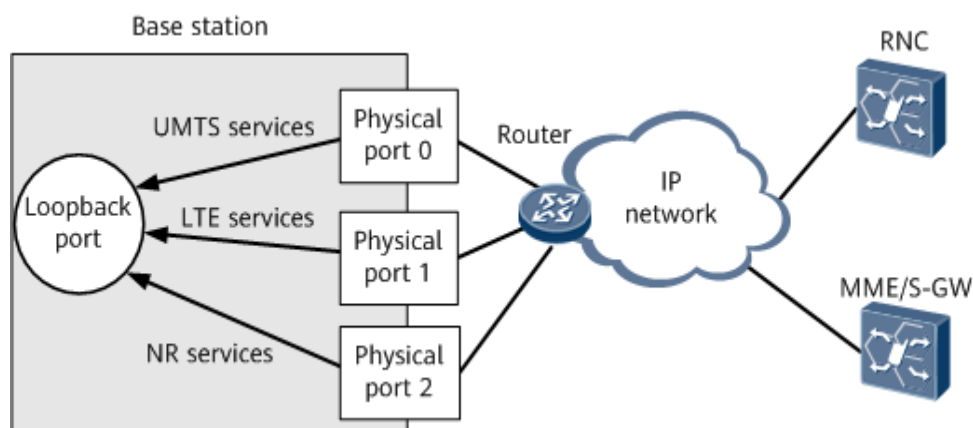


Figure 4-4 Scenario 2



Scenario 2 is not a multimode base station co-transmission networking scenario. In this scenario, it is recommended that the **DLFLOWCTRLPARA.FAIRSWITCH** (in the old model)/**IPDLFLOWCTRLPARA.FAIRSWITCH** (in the new model) parameter be configured for the loopback port.

The fair flow control switch ensures that at least 30% of the actual receive bandwidth is retained for UMTS HSDPA services. This does not mean that the bandwidth retained for UMTS HSDPA services is equal to 30% of the actual receive bandwidth.

When multiple physical ports are used for load sharing in co-transmission scenarios, flow control can still take effect but cannot be precisely implemented.

For details about the flow control algorithm, see *Transmission Resource Management* in *RAN Feature Documentation*.

4.1.4 Application Scenarios

4.1.4.1 Unlimited Access Bandwidth for Multimode Base Stations

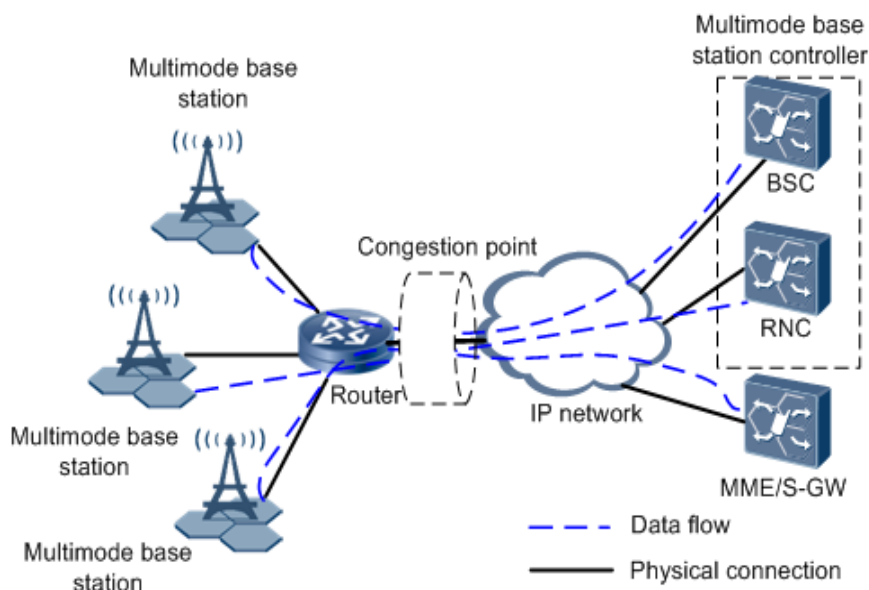
4.1.4.1.1 Introduction

Access bandwidth unlimited for multimode base stations refers to scenarios in which:

- The operator cannot or has not planned access bandwidth for each multimode base station.
- The bandwidth of the converging device, which converges the data of multimode base stations, is either limited or unlimited.

For example, in [Figure 4-5](#), the access bandwidth for each multimode base station is 100 Mbit/s and the bandwidth for intermediate transmission devices is also 100 Mbit/s.

Figure 4-5 Access bandwidth unlimited for multimode base stations



4.1.4.1.2 Transmission Resource Management Policy Configuration

(Optional) Configuring Traffic Limiting and Shaping on the Base Station Controller Side

Traffic limiting and shaping can be configured on the base station controller side if the operator can estimate the transmission bandwidth required by a base station based on the traffic model. The bandwidth after rate limitation is calculated based on the service model.

Configuring the Mapping Between Traffic Classes and Transmission Priorities

For details about the recommended transmission priority mapping for each traffic class of GSM, UMTS, LTE, and NR services, see *Transmission Resource Management* for GSM BSS, WCDMA RAN, eRAN, and 5G RAN, respectively. For details about how to configure the mapping between DSCP values and traffic classes for various RATs, see [4.1.3.1 Transmission Priorities](#).

Generally, network devices support queue scheduling. Layer 3 and Layer 2 network devices support eight priority queues. However, if network devices on the bearer network support less than eight queues, transmission priority combining policies listed in [Table 4-6](#) are recommended. You can combine packets with different DSCP values or VLAN priorities into one queue. For example, if there are six queues, packets with DSCP values 48 and 46 are placed in the same queue, and packets with VLAN priorities 6 and 5 are placed in the same queue, and this queue has the highest priority.

Table 4-6 Recommended transmission priority combining policies (access bandwidth unlimited for multimode base stations)

Number of Queues	DSCP Value Combining Policy	VLAN Priority Combing Policy
6	DSCP values for the six queues are (48+46), 34, 26, 18, 10, and 0.	VLAN priorities for the six queues are (6+5), 4, 3, 2, 1, and 0.
5	DSCP values for the five queues are (48+46), (34+26), 18, 10, and 0.	VLAN priorities for the five queues are (6+5), (4+3), 2, 1, and 0.
4	DSCP values for the four queues are (48+46), (34+26+18), 10, and 0.	VLAN priorities for the four queues are (6+5), (4+3+2), 1, and 0.
3	DSCP values for the three queues are (48+46), (34+26+18+10), and 0.	VLAN priorities for the three queues are (6+5), (4+3+2+1), and 0.

If there are only two queues, contact technical support engineers to confirm the DSCP value combining policy.

Configuring the Transmission Flow Control Algorithm Switch

Table 4-7 lists the recommended settings of the dynamic flow control algorithm switch and fair flow control switch of the NodeB. Set **ULFLOWCTRLPARA.TNLCONGCTRLSWITCH** (in the old model)/**IPULFLOWCTRLPARA.TNLCONGCTRLSWITCH** (in the new model), **DLFLOWCTRLPARA.SWITCH** (in the old model)/**IPDLFLOWCTRLPARA.SWITCH** (in the new model), and **DLFLOWCTRLPARA.FAIRSWITCH** (in the old model)/**IPDLFLOWCTRLPARA.FAIRSWITCH** (in the new model).

Table 4-7 Recommended settings for the NodeB flow control algorithm switch (access bandwidth unlimited for multimode base stations)

Base Station Type	Congestion Control Switch	Flow Control Switch	Fair Flow Control Switch
Separate-MPT GU dual-mode base station	Retain the default value.	Retain the default value.	N/A
Co-MPT GU dual-mode base station			
GL/GT/GN/GLT/GLN/GTN/LN/TN/LT/LTN/GLTN separate-MPT multimode base station	N/A	N/A	N/A
GL/GT/GN/GLT/GLN/GTN/LN/TN/LT/LTN/GLTN co-MPT multimode base station			
UL/UT/UN/GUL/GUT/GUN/ULT/UTN/ULN/GULT/GULN/GUTN/ULTN/GULTN separate-MPT multimode base station	Retain the default value.	Retain the default value. The bearer network must support two or more queues.	Set this parameter to ON .
UL/UT/UN/GUL/GUT/GUN/ULT/UTN/ULN/GULT/GULN/GUTN/ULTN/GULTN co-MPT multimode base station			

4.1.4.2 Limited Access Bandwidth for Multimode Base Stations

4.1.4.2.1 Introduction

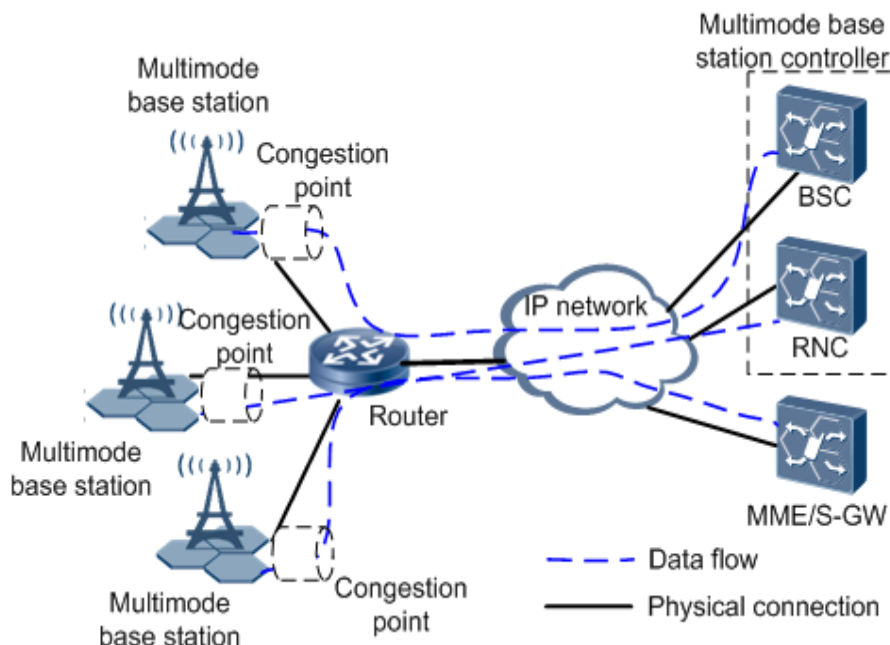
Limited access bandwidth for multimode base stations refers to scenarios in which:

- The maximum data rate for each multimode base station must not exceed the planned bandwidth.
- The bandwidth of intermediate transmission devices is either limited or unlimited.

The access bandwidth for a base station is limited if the bearer network is leased or if the base station uses satellite, microwave, or xPON to receive data.

For example, in [Figure 4-6](#), the access bandwidth for the three multimode base stations is limited to 10 Mbit/s.

Figure 4-6 Limited access bandwidth for multimode base stations



4.1.4.2.2 Transmission Resource Management Strategies

Configuring Traffic Limiting and Shaping on the Base Station Controller Side

Configure traffic limiting and shaping on the base station controller side and set the bandwidth limit to the access bandwidth planned by the operator for a multimode base station.

Configuring Traffic Limiting and Shaping on the Co-Transmission Port of the Base Station Side

Configure traffic limiting and shaping on the co-transmission port of the base station side and set the bandwidth limit to the access bandwidth planned by the operator for a multimode base station.

Configuring the Mapping Between Traffic Classes and DSCP Values

For details about the recommended transmission priority mapping for each traffic class of GSM, UMTS, LTE, and NR services, see *Transmission Resource Management* for GSM BSS, WCDMA RAN, eRAN, and 5G RAN, respectively.

In most cases, intermediate transmission devices support queue scheduling. Layer 3 and Layer 2 intermediate transmission devices support eight queues. However, if intermediate transmission devices in the bearer network support less than eight queues, transmission priority combining strategies listed in [Table 4-8](#) are recommended. You can combine packets with different DSCP values into one queue and combine packets with different VLAN priorities into one queue. For example, if the intermediate transmission devices support six queues, packets

whose DSCP values are 48 and 46 can be put into one queue. Accordingly, packets whose VLAN priorities are 6 and 5 can be put into one queue. This queue has the highest transmission priority.

Table 4-8 Recommended transmission priority combining strategies if access bandwidth is limited for multimode base stations

Number of Queues	DSCP Value for Each Queue	VLAN Priority for Each Queue
6	(48+46), 34, 26, 18, 10, 0	(6+5), 4, 3, 2, 1, 0
5	(48+46), (34+26), 18, 10, 0	(6+5), 4, 3, 2, (1+0)
4	(48+46), (34+26+18), 10, 0	(6+5), (4+3+2), 1, 0
3	(48+46), (34+26+18+10), 0	(6+5), (4+3+2+1), 0

If there are only two queues, obtain from Huawei technical support personnel the method of combining DSCP values.

Configuring the Flow Control Algorithm

Table 4-9 provides recommended settings for the NodeB flow control algorithm and the HSDPA fair flow control switch. Pay attention to the following in data configurations:

- The **RSCGRPALG.TCSW** (in the old model)/**IPRSCGRPALG.TCSW** (in the new model) parameter is set to **ENABLE** by default. If you want to set this parameter to **DISABLE**, first add a default transmission resource group to the co-transmission port. Then set **RSCGRPALG.TCSW** (in the old model)/**IPRSCGRPALG.TCSW** to **DISABLE** for the default transmission resource group you have added.

If the co-transmission port is configured with a transmission resource group whose **RSCGRP.RSCGRPID** (in the old model)/**IPRSCGRP.IPRSCGRPID** (in the new model) is set to **AUTOPORT**, first run the **RMV RSCGRP** (in the old model)/**RMV IPRSCGRP** (in the new model) command to delete this group and then run the **ADD RSCGRP** (in the old model)/**ADD IPRSCGRP** (in the new model) command with **RSCGRP.RSCGRPID** (in the old model)/**IPRSCGRP.IPRSCGRPID** (in the new model) set to **DEFAULTPORT** to add a transmission resource group.

If the co-transmission port is not configured with a transmission resource group, run the **ADD RSCGRP** (in the old model)/**ADD IPRSCGRP** (in the new model) command with **RSCGRP.RSCGRPID** (in the old model)/**IPRSCGRP.IPRSCGRPID** (in the new model) set to **DEFAULTPORT** to add a transmission resource group.

- If the configured UL CIR is less than or equal to 2 Mbit/s, you need to set **RSCGRPALG.TCSW** (in the old model)/**IPRSCGRPALG.TCSW** (in the new model) to **DISABLE**. Otherwise, the OM service rate may decrease dramatically.

Table 4-9 Recommended settings for the NodeB flow control algorithm and the HSDPA fair flow control switch in the case of limited access bandwidth for multimode base stations

Base Station Type	Setting of RSCGRPALG.TCSW (in the Old Model)/IPRSCGRPALG.TCSW (in the New Model)	Setting of ULFLOWCTRLP.ARA.TNLCONGCTRLSWITCH (in the Old Model)/IPULFLOWCTRLP.ARA.TNLCONGCTRLSWITCH (in the New Model)	Setting of DLFLOWCTRLP.ARA.SWITCH (in the Old Model)/IPDLFLOWCTRLP.ARA.SWITCH (in the New Model)	Setting of DLFLOWCTRLP.ARA.FAIRSWITCH (in the Old Model)/IPDLFLOWCTRLP.ARA.FAIRSWITCH (in the New Model)
Separate-MPT GU dual-mode base station	ENABLE(Enable) (default value)	ON(On) (default value)	BW_SHAPING_ONOFF_TOGGLE(BW_SHAPING_ONOFF_TOGGLE) (default value)	N/A
Co-MPT GU dual-mode base station				
GL/GT/GN/GLT/GLN/GTN/LN/TN/LT/LTN/GLTN separate-MPT multimode base station	ENABLE(Enable) (default value)	N/A	N/A	N/A
GL/GT/GN/GLT/GLN/GTN/LN/TN/LT/LTN/GLTN co-MPT multimode base station				
UL/UT/UN/GUL/GUT/GUN/ULT/UTN/ULN/GULT/GULN/GUTN/ULTN/GULTN separate-MPT multimode base station	ENABLE(Enable) (default value): if co-transmission is implemented through backplane interconnection DISABLE(Disable) (default value): if co-transmission is implemented through panel interconnection	ON(On) (default value)	BW_SHAPING_ONOFF_TOGGLE(BW_SHAPING_ONOFF_TOGGLE) (default value) The bearer network must support two or more queues.	ENABLE

Base Station Type	Setting of RSCGRPALG.7CSW (in the Old Model)/IPRSCGRPALG.7CSW (in the New Model)	Setting of ULFLOWCTRLP.ARA.TNLCONGCTRLSWITCH (in the Old Model)/IPULFLOWCTRLP.ARA.TNLCONGCTRLSWITCH (in the New Model)	Setting of DLFLOWCTRLP.ARA.SWITCH (in the Old Model)/IPDLFLOWCTRLP.ARA.SWITCH (in the New Model)	Setting of DLFLOWCTRLP.ARA.FAIRSWITCH (in the Old Model)/IPDLFLOWCTRLP.ARA.FAIRSWITCH (in the New Model)
UL/UT/UN/GUL/GUT/GUN/ULT/UTN/ULN/GULT/GULN/GUTN/ULTN/GULTN co-MPT multimode base station	ENABLE(Enable) (default value)	ON(On) (default value)	BW_SHAPING_ONOFF_TOGGLE(BW_SHAPING_ONOFF_TOGGLE) (default value) The bearer network must support two or more queues.	ENABLE

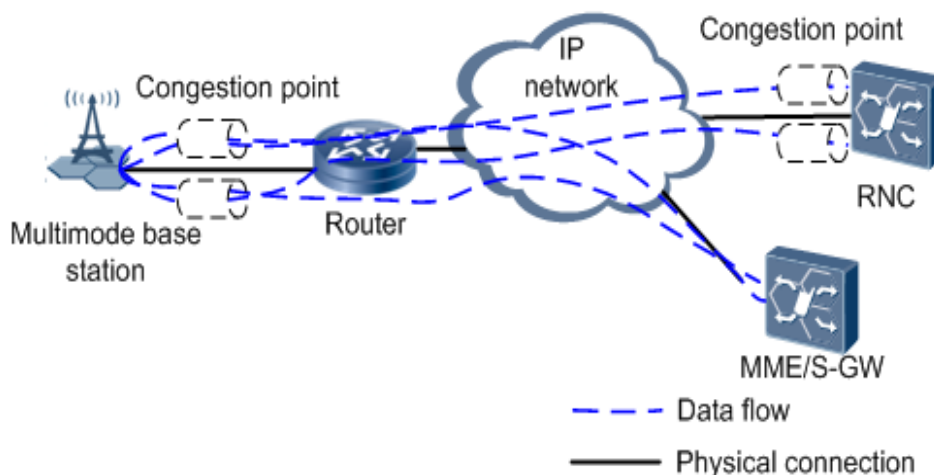
Configuring the Load Control Algorithm

When co-transmission is applied, the load control algorithm for each mode in a multimode base station is configured in the same way as the load control algorithm in a single-mode base station. For details about load control for GSM, UMTS, LTE, and NR, see *Transmission Resource Management* for GSM BSS, WCDMA RAN, eRAN, and 5G RAN, respectively.

4.1.4.3 Limited Access Bandwidth for Each Operator in RAN Sharing Scenarios

4.1.4.3.1 Introduction

Access bandwidth limited in radio access network (RAN) sharing scenarios refer to scenarios in which: multiple operators share one multimode base station. Access bandwidth of one operator cannot be shared with other operators. Instead, the access bandwidth of this operator can be shared among services of each mode run by this operator. The access bandwidth for each operator is limited. Access bandwidth for each operator is limited when the bearer network is a leased one. In the current version, limited access bandwidth for multiple operators in RAN sharing scenarios applies only to UL dual-mode base stations. For example, in [Figure 4-7](#), the access bandwidth for each operator is limited to 10 Mbit/s.

Figure 4-7 Limited access bandwidth for each operator in RAN sharing scenarios

4.1.4.3.2 Transmission Resource Management Strategies

Configuring Traffic Limiting and Shaping on the Base Station Controller Side

Configure a logical port for each operator on the base station controller side. Set the bandwidth limited on the logical port to the access bandwidth planned by the operator.

Configuring Traffic Limiting and Shaping on the Co-Transmission Port of the Base Station Side

Configure a logical port for each operator on the co-transmission port of the base station side. Set the bandwidth limited on the logical port to the access bandwidth planned by the operator.

Configuring the Mapping Between Traffic Classes and DSCP Values

For details about the recommended transmission priority mapping for each traffic class of GSM, UMTS, LTE, and NR services, see *Transmission Resource Management* for GSM BSS, WCDMA RAN, eRAN, and 5G RAN, respectively.

In most cases, intermediate transmission devices support queue scheduling. Layer 3 and Layer 2 intermediate transmission devices support eight queues. However, if intermediate transmission devices in the bearer network support less than eight queues, transmission priority combining strategies listed in [Table 4-10](#) are recommended. You can combine packets with different DSCP values into one queue and combine packets with different VLAN priorities into one queue. For example, if the intermediate transmission devices support six queues, packets whose DSCP values are 48 and 46 can be put into one queue. Accordingly, packets whose VLAN priorities are 6 and 5 can be put into one queue. This queue has the highest transmission priority.

Table 4-10 Recommended transmission priority combining strategies if access bandwidth is limited for each operator in RAN sharing scenarios

Number of Queues	DSCP Value for Each Queue	VLAN Priority for Each Queue
6	(48+46), 34, 26, 18, 10, 0	(6+5), 4, 3, 2, 1, 0
5	(48+46), (34+26), 18, 10, 0	(6+5), 4, 3, 2, (1+0)
4	(48+46), (34+26+18), 10, 0	(6+5), (4+3+2), 1, 0
3	(48+46), (34+26+18+10), 0	(6+5), (4+3+2+1), 0

If there are only two queues, obtain from Huawei technical support personnel the method of combining DSCP values.

Configuring the Flow Control Algorithm

[Table 4-11](#) lists the recommended configurations. The configuration notes are the same as those of [Configuring the Flow Control Algorithm](#) in [4.1.4.2.2 Transmission Resource Management Strategies](#).

Table 4-11 Recommended settings for the NodeB flow control algorithm and the HSDPA fair flow control switch if access bandwidth is limited for each operator in RAN sharing scenarios

Scenario	Setting of RSCGRPALG.TCSW (in the Old Model)/IPRSCGRPALG.TCSW (in the New Model)	Setting of ULFLOWCTRLP.ARA.TNLCONGCTRLSWITCH (in the Old Model)/IPULFLOWCTRLP.ARA.TNLCONGCTRLSWITCH (LTE eNodeB, 5G gNodeB) (in the New model)	Setting of DLFLOWCTRLP.ARA.SWITCH (in the Old Model)/IPDLFLOWCTRLP.ARA.SWITCH (in the New Model)	Setting of DLFLOWCTRLP.ARA.FAIRSWITCH (in the Old Model)/DLFLOWCTRLP.ARA.FAIRSWITCH (in the New Model)
UL/UT/UN/GUL/GUT/GUN/ULT/UTN/ULN/GULT/GULN/GUTN/ULTN/GULTN separate-MPT multimode base station	- if co-transmission is implemented through backplane interconnection ENABLE(Enable) (default value) DISABLE(Disable): if co-transmission is implemented through panel interconnection	ON(default value)	BW_SHAPING_ONOFF_TOGGLE (default value) The bearer network must support two or more queues.	ENABLE
UL/UT/UN/GUL/GUT/GUN/ULT/UTN/ULN/GULT/GULN/GUTN/ULTN/GULTN co-MPT multimode base station	ENABLE (default value)			

Configuring the Load Control Algorithm

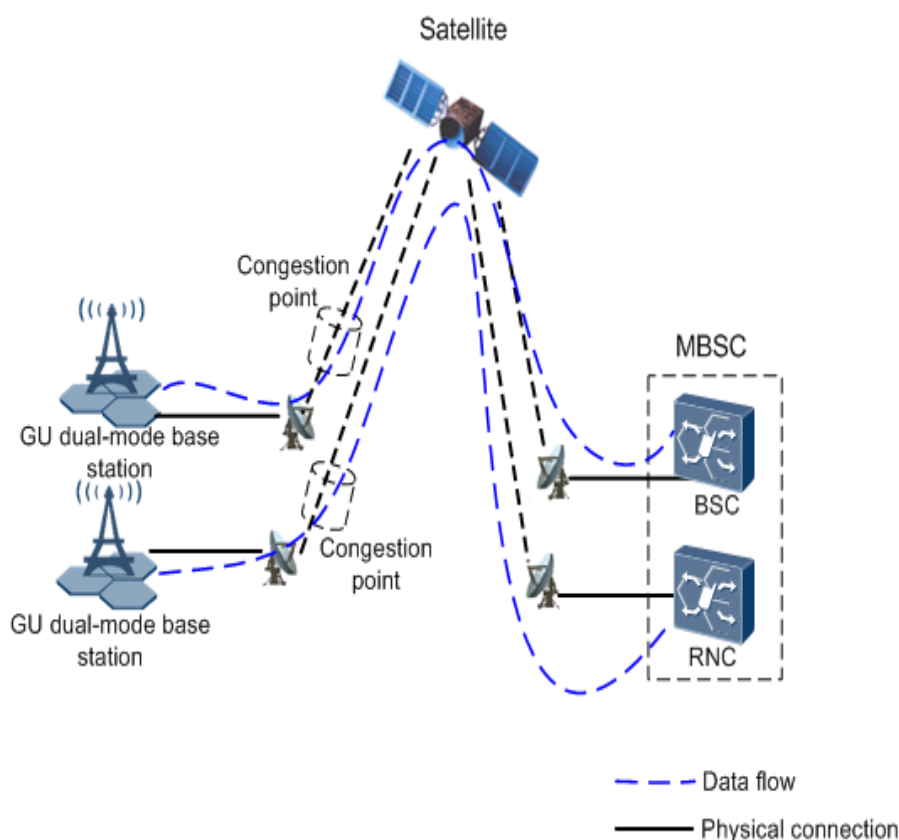
When co-transmission is applied, the load control algorithm for each mode in a multimode base station is configured in the same way as the load control algorithm in a single-mode base station. For detailed description about GSM, UMTS, LTE, and NR, see *Transmission Resource Management* for GSM BSS, WCDMA RAN, eRAN, and 5G RAN, respectively.

4.1.4.4 Satellite Transmission for Multimode Base Stations (Without Distinguishing Transmission Priorities)

4.1.4.4.1 Introduction

Satellite transmission for multimode base stations (without distinguishing transmission priorities) refers to scenarios in which differentiated services cannot be performed on satellite transmission networks based on their transmission priorities. This scenario is applicable only to GU dual-mode base stations. **Figure 4-8** uses an example of two GU dual-mode base stations for illustrating satellite transmission. The access bandwidth for each GU dual-mode base station is limited to 4 Mbit/s and the satellite transmission network where the GU dual-mode base stations access cannot distinguish their transmission priorities. In this case, transmission packets are processed according to the first-come first-served policy.

Figure 4-8 Satellite transmission for GU dual-mode base stations



4.1.4.4.2 Transmission Resource Management Strategies

Configuring Traffic Limiting and Shaping on the Base Station Controller Side

Configure traffic limiting and shaping on the base station controller side. The bandwidth after rate limitation for GSM mode is the access bandwidth for multimode base stations planned by the operator. The bandwidth after rate limitation for UMTS mode is the access bandwidth for UMTS planned by the operator. The access bandwidth for UMTS mode must be less than that for

multimode base stations, so that the access bandwidth for UMTS mode cannot preempt that for GSM mode.

Configuring Traffic Limiting and Shaping on the Co-Transmission Port of the Base Station Side

Configure traffic limiting and shaping on the co-transmission port of the base station side.

Configuring the Mapping Between Traffic Classes and DSCP Values

For details about the recommended transmission priority mapping for each traffic class of GSM, UMTS, LTE, and NR services, see *Transmission Resource Management* for GSM BSS, WCDMA RAN, eRAN, and 5G RAN, respectively.

In this scenario, the satellite transmission does not distinguish transmission priorities. Therefore, the transmission priority for each traffic class takes effect on the radio equipment side.

Configuring the Flow Control Algorithm

Table 4-12 provides recommended settings for the NodeB flow control algorithm.

Table 4-12 Recommended settings for the NodeB flow control algorithm if satellite transmission is used for GU dual-mode base stations

Base Station Type	Setting of the RSCGRPALG.TCSW (in the Old Model)/ IPRSCGRPALG.TCSW (in the New Model)	Setting of ULFLOWCTRL PARA.TNLCONGCTRLSWITCH (in the Old Model)/ IPULFLOWCTRLPARA.TNLCONGCTRLSWITCH (LTE eNodeB, 5G gNodeB) (in the New Model)	Setting of DLFLOWCTRLPARA.SWITCH (in the Old Model)/ IPDLFLOWCTRLPARA.SWITCH (in the New Model)	Setting of DLFLOWCTRLPARA.FAIRSWITCH (in the Old Model)/ IPDLFLOWCTRLPARA.FAIRSWITCH (in the New Model)
Separate-MPT GU dual-mode base station	ENABLE(Enable) (default value)	ON(On) (default value)	BW_SHAPING_ONOFF_TOGGLE(BW_SHAPING_ONOFF_TOGGLE) (default value)	N/A
Co-MPT GU dual-mode base station				

During information configuration of the physical port, if the configured UL CIR is less than or equal to 2 Mbit/s, you need to set **RSCGRPALG.TCSW** (in the old

model)/**IPRSCGRPALG.TCSW** (in the new model) to **DISABLE**. Otherwise, the OM service rate may decrease dramatically.

Configuring the Load Control Algorithm

When co-transmission is applied, the load control algorithm for each mode in a multimode base station is configured in the same way as the load control algorithm in a single-mode base station. For details about these policies on the GSM and UMTS sides, see *Transmission Resource Management* in *GBSS Feature Documentation* and *RAN Feature Documentation*, respectively.

4.2 Network Analysis

4.2.1 Benefits

GSM, UMTS, LTE, and NR services have different peak hours. Therefore, transmission resources of one RAT can be multiplexed by other RATs if this RAT is not experiencing a traffic peak. The Bandwidth Sharing of Multimode Base Station Co-Transmission feature provides the following benefits:

- For a multimode base station in co-transmission scenarios, transmission resources can be dynamically shared among each RAT of the multimode base station.
- As GSM services continuously shrink, the released GSM bandwidth can be used by NR, LTE and UMTS services.
- When uplink or downlink transmission resources of a multimode base station are congested, the quality of service (QoS) of high-priority GSM, UMTS, LTE, and NR services can be guaranteed.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

If the settings of inter-RAT parameters, such as inter-RAT bandwidth allocation and inter-RAT QoS planning, are inappropriate, activating this feature will have the following impacts:

- Increased service congestion rates
- Reduced data rates of low-priority services, for example, best effort (BE) services
- Increased packet loss rates of low-priority services

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

- MRFD-121115 Bandwidth sharing of MBTS Multi-mode Co-Transmission(GBTS)

RAT	Function Name	Function Switch	Reference
GSM	IP-Based Multi-mode Co-Transmission on BS side(GBTS)	None	<i>Common Transmission</i>

- MRFD-121125 Bandwidth sharing of MBTS Multi-mode Co-Transmission(NodeB)

RAT	Function Name	Function Switch	Reference
UMTS	IP-Based Multi-mode Co-Transmission on BS side(NodeB)	None	<i>Common Transmission</i>

- MRFD-121135 Bandwidth sharing of MBTS Multi-mode Co-Transmission(eNodeB)

RAT	Function Name	Function Switch	Reference
LTE FDD	IP-Based Multi-mode Co-Transmission on BS side(eNodeB)	None	<i>Common Transmission</i>

- MRFD-121145 Bandwidth sharing of MBTS Multi-mode Co-Transmission(LTE TDD)

RAT	Function Name	Function Switch	Reference
LTE TDD	IP-Based Multi-mode Co-Transmission on BS side(LTE TDD)	None	<i>Common Transmission</i>

- MRFD-121155 Bandwidth sharing of MBTS Multi-mode Co-Transmission(NB-IoT)

RAT	Function Name	Function Switch	Reference
NB-IoT	IP-Based Multi-mode Co-Transmission on BS side(NB-IoT)	None	<i>Common Transmission</i>

- MRFD-151168 Bandwidth sharing of MBTS Multi-mode Co-Transmission(NR)

RAT	Function Name	Function Switch	Reference
NR FDD Low-frequency NR TDD	IP-Based Multi-mode Co-Transmission on BS side(NR)	None	<i>Common Transmission</i>

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

RAT	Base Station Model
GSM	3900 and 5900 series base stations
UMTS	3900 and 5900 series base stations DBS3900 LampSite and DBS5900 LampSite
LTE	3900 and 5900 series base stations (macro base stations) DBS3900 LampSite and DBS5900 LampSite
NR	3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910. - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

No requirements

RF Modules

N/A

4.3.4 Networking

- Transmission bandwidth plan for radio services
Create a transmission bandwidth plan each for GSM, UMTS, LTE, and NR of a multimode base station based on the service plan and corresponding bandwidth requirements.
- QoS plan for radio services
For a multimode base station in co-transmission scenarios, it is recommended that signaling and circuit switched (CS) services be classified as real-time services and packet switched (PS) services as non-real-time services for GSM, UMTS, LTE, and NR. Set real-time services to a higher priority than non-real-time services to ensure the continuity of signaling and CS services when transmission resources become congested. Activate the flow control algorithm for each RAT to properly allocate transmission resources across non-real-time services when transmission resources become congested.
- Mapping between traffic classes and transmission priorities
Plan traffic classes, DSCP values, VLAN priorities, and the mapping between traffic classes and DSCP values based on the QoS plan of services.
- QoS plan for the bearer network
Plan DSCP values for Layer-3 devices, and plan VLAN priorities and DSCP values for Layer-2 devices based on service priorities.
- Bandwidth plan for the bearer network
Plan bandwidth for the bearer network based on services' bandwidth requirements and available bandwidth resources. When planning transmission bandwidth on the RAN side, ensure that the bandwidth between a base station and a base station controller is higher than the total bandwidth of real-time services. This guarantees the service quality of real-time services.

4.3.5 Others

To provide guide on how to plan transmission bandwidth and transmission priorities for multimode base stations and multimode base station controllers, you need to know the network topology and transmission bandwidth plan of the bearer network, which include transmission bandwidth available in the bearer network and queues available on transmission devices. If bandwidth resources across the RATs are inappropriately allocated, reallocate the bandwidth resources based on the traffic model.

To implement the Bandwidth Sharing of Multimode Base Station Co-Transmission feature, the bearer network must support QoS management. Otherwise, this

feature becomes invalid when the bearer network is congested. QoS management includes the following aspects:

- Layer-3 devices support DSCP-priority-based QoS management.
- Layer-2 devices support VLAN-priority-based QoS management.
- Intermediate transmission devices support the PQ+WRR queue scheduling function and at least two PQ queues are supported. (WRR stands for weighted round robin.)

4.4 Operation and Maintenance

4.4.1 When to Use

It is recommended that this feature be activated for a multimode base station where IP-based co-transmission is applied. By doing this, bandwidth resources can be appropriately allocated across all the RATs of the multimode base station.

4.4.2 Data Configuration

4.4.2.1 Data Preparation

Traffic Limiting and Shaping

If access bandwidth is limited for multimode base stations, data for traffic limiting and shaping must be prepared on the base station side that provides a co-transmission port. The following table lists the key data that must be set in an **LR** (in the old model)/**PORTLR** (in the new model) MO to configure traffic limiting and shaping.

When **GTRANSPARA.TRANS CFGMODE** is set to **OLD**:

Table 4-13 Data to prepare for configuring traffic limiting and shaping if access bandwidth is limited for multimode base stations

Parameter Name	Parameter ID	Setting Notes
UL Committed Information Rate	LR.CIR	Set these parameters based on the network plan.
Committed Burst Size	LR.CBS	
Excess Burst Size	LR.EBS	

When **GTRANSPARA.TRANS CFGMODE** is set to **NEW**:

Table 4-14 Data to prepare for configuring traffic limiting and shaping if access bandwidth is limited for multimode base stations

Parameter Name	Parameter ID	Setting Notes
UL Committed Information Rate	PORTLR.CIR	Set these parameters based on the network plan.
Committed Burst Size	PORTLR.CBS	
Excessive Burst Size	PORTLR.EBS	

If access bandwidth is limited for each operator in multi-operator scenarios, data for traffic limiting and shaping must be prepared on the base station side that provides a co-transmission port. The following table lists the key data that must be set in an **RSCGRP** (in the old model)/**IPRSCGRP** (in the new model) MO to configure traffic limiting and shaping.

When **GTRANSPARA.TRANS CFGMODE** is set to **OLD**:

Table 4-15 Data to prepare for configuring traffic limiting and shaping if access bandwidth is limited for each operator in multi-operator scenarios

Parameter Name	Parameter ID	Setting Notes
Tx Bandwidth	RSCGRP.TXBW	Set these parameters based on the network plan.
TX Committed Burst Size	RSCGRP.TXCBS	
TX Excessive Burst Size	RSCGRP.TXEBS	

When **GTRANSPARA.TRANS CFGMODE** is set to **NEW**:

Table 4-16 Data to prepare for configuring traffic limiting and shaping if access bandwidth is limited for each operator in multi-operator scenarios

Parameter Name	Parameter ID	Setting Notes
TX Bandwidth	IPRSCGRP.TXBW	Set these parameters based on the network plan.
TX Committed Burst Size	IPRSCGRP.TXCBS	
TX Excessive Burst Size	IPRSCGRP.TXEBS	

Table 4-17 lists the optional data to prepare for configuring traffic limiting and shaping on the GBSC or RNC side if access bandwidth is unlimited for multimode

base stations or access bandwidth is limited for each operator in multi-operator scenarios.

Table 4-17 Data to prepare for configuring traffic limiting and shaping on the GBSC or RNC side

Parameter Name	Parameter ID	Setting Notes
Logic Port No.	IPLOGICPORT.LPN	Set this parameter to the number of the GBSC/RNC logical port.
Bandwidth [64kbps]	IPLOGICPORT.CIR	Set this parameter to the access bandwidth planned by the operator or bandwidth calculated by the traffic model. When the access bandwidth is limited for each operator in multi-operator scenarios, set this parameter to the access bandwidth planned by each operator.

Transport QoS

- Transport QoS for GSM services
 - **Table 4-18** lists the data to prepare for configuring the mapping between management-, control-, and user-plane data and DSCP values for the GBTS.
 - **Table 4-19** lists the data to prepare for configuring the mapping between management- and control-plane data and DSCP values for the eGBTS.
 - **Table 4-22** lists the data to prepare for configuring the mapping between management- and control-plane data and DSCP values for the GBSC. **Table 4-23** lists the data to prepare for configuring the mapping between user-plane data and DSCP values for the GBSC.
- Transport QoS for UMTS services
 - **Table 4-19** lists the data to prepare for configuring the mapping between management- and control-plane data and DSCP values for the NodeB.
 - **Table 4-24** lists the data to prepare for configuring the mapping between management-, control-, and user-plane data and DSCP values for the RNC.
- Transport QoS for LTE services

Table 4-19 lists the data to prepare for configuring the mapping between management- and control-plane data and DSCP values for the eNodeB. **Table 4-21** lists the data to prepare for configuring the mapping between user-plane data and DSCP values for the eNodeB.
- Transport QoS for NR services

Table 4-19 lists the data to prepare for configuring the mapping between management- and control-plane data and DSCP values for the gNodeB. **Table**

4-21 lists the data to prepare for configuring the mapping between user-plane data and DSCP values for the gNodeB.

Table 4-18 Data to prepare for configuring the mapping between management-, control-, and user-plane data and DSCP values for the GBTS

Parameter Name	Parameter ID	Setting Notes
Service Type	BTSVLAN.SERVICETY PE	See the recommended parameter configurations in 4.1.4 Application Scenarios .
DSCP	BTSVLAN.DSCP	

Table 4-19 Data to prepare for configuring the mapping between the management- and control-plane data and DSCP values for the eGBTS/NodeB/eNodeB/gNodeB/co-MPT multimode base station

Parameter Name	Parameter ID	Setting Notes
Priority Rule	DIFPRI.PRIRULE	Set this parameter to DSCP(DSCP) .
Signaling Priority	DIFPRI.SIGPRI	See the recommended parameter configurations in 4.1.4 Application Scenarios .
OM High Priority	DIFPRI.OMHIGHPRI	
OM Low Priority	DIFPRI.OMLOWPRI	
IP Clock Priority	DIFPRI.IPCLKPRI	

Table 4-20 Data to prepare for configuring the RAT-specific mapping between control-plane data and DSCP values for the eGBTS/NodeB/eNodeB/gNodeB/co-MPT multimode base station

Parameter Name	Parameter ID	Setting Notes
DSCP Switch	SCTPLNK.DSCPSW	Set this parameter based on the network plan.
DSCP	SCTPLNK.DSCP	Set this parameter to the default value. If you need to change the value, set this parameter based on the network plan.
DSCP Switch	SCTPTEMPLATE.DSCP SW	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
DSCP	SCTPTEMPLATE.DSCP	Set this parameter to the default value. If you need to change the value, set this parameter based on the network plan.

Table 4-21 Data to prepare for configuring the mapping between user-plane data and DSCP values for the eNodeB/NR side

Parameter Name	Parameter ID	Setting Notes
User Data Type Transfer Parameter Group ID	UDTPARAGRP.UDTPARAGRPID	Set this parameter to a value ranging from 40 to 48 for user data types 1 to 9.
Priority	UDTPARAGRP.PRI	See the recommended parameter configurations in 4.1.4 Application Scenarios .

Table 4-22 Data to prepare for configuring the mapping between management- and control-plane data and DSCP values for the GBSC

Parameter Name	Parameter ID	Setting Notes
OML DSCP	BSCABISPRIMAP.OML DSCP	See the recommended parameter configurations in 4.1.4 Application Scenarios .
RSL DSCP	BSCABISPRIMAP.RSL DSCP	
EML DSCP	BSCABISPRIMAP.EML DSCP	
ESL DSCP	BSCABISPRIMAP.ESL DSCP	

Table 4-23 Data to prepare for configuring the mapping between the user-plane data and DSCP values for the GBSC

Parameter Name	Parameter ID	Setting Notes
CS voice path	TRMMAP.CSVOICEPATH	See the recommended parameter configurations in 4.1.4 Application Scenarios .
CS data path	TRMMAP.CSDATAPATH	

Parameter Name	Parameter ID	Setting Notes
PS high PRI data path	TRMMAP.PSHPRIDAT APATH	
PS low PRI data path	TRMMAP.PSLPRIDAT APATH	

Table 4-24 Data to prepare for configuring the mapping between the control- and user-plane data and DSCP values for the RNC

Parameter Name	Parameter ID	Setting Notes
Common channel primary path	TRMMAP.CCHPRIPATH	See the recommended parameter configurations in 4.1.4 Application Scenarios .
IMS SRB primary path	TRMMAP.SIPPRIPATH	
SRB primary path	TRMMAP.SRBPRIPATH	
AMR voice primary path	TRMMAP.VOICEPRIPATH	
R99 CS conversational primary path	TRMMAP.CSCONVPRI PATH	
R99 CS streaming primary path	TRMMAP.CSSTRMPRI PATH	
R99 PS conversational primary path	TRMMAP.PSCONVPRI PATH	
R99 PS streaming primary path	TRMMAP.PSSTRMPRI PATH	
R99 PS high PRI interactive primary path	TRMMAP.PSINTHGHP RIPATH	
R99 PS middle PRI interactive primary path	TRMMAP.PSINTMIDP RIPATH	
R99 PS low PRI interactive primary path	TRMMAP.PSINTLOW PRIPATH	
R99 PS background primary path	TRMMAP.PSBKGPRIPATH	
HSDPA Signal primary path	TRMMAP.HDSRBPRIPATH	

Parameter Name	Parameter ID	Setting Notes
HSDPA IMS Signal primary path	TRMMAP.HDSIPPRIPATH	
HSDPA Voice primary path	TRMMAP.HDVOICEPRIPATH	
HSDPA conversational primary path	TRMMAP.HDCONVPRIPATH	
HSDPA streaming primary path	TRMMAP.HDSTRMPRIPATH	
HSDPA high PRI interactive primary path	TRMMAP.HDINTHGHPRIPATH	
HSDPA middle PRI interactive primary path	TRMMAP.HDINTMIDPRIPATH	
HSDPA low PRI interactive primary path	TRMMAP.HDINTLOWPRIPATH	
HSDPA background primary path	TRMMAP.HDBKGPRIPATH	
HSUPA Signal primary path	TRMMAP.HUSRBPRIPATH	
HSUPA IMS Signal primary path	TRMMAP.HUSIPPRIPATH	
HSUPA voice primary path	TRMMAP.HUVOICEPRIPATH	
HSUPA conversational primary path	TRMMAP.HUCONVPRIPATH	
HSUPA streaming primary path	TRMMAP.HUSTRMPRIPATH	
HSUPA high PRI interactive primary path	TRMMAP.HUINTHGHPRIPATH	
HSUPA middle PRI interactive primary path	TRMMAP.HUINTMIDPRIPATH	
HSUPA low PRI interactive primary path	TRMMAP.HUINTLOWPRIPATH	

Parameter Name	Parameter ID	Setting Notes
HSUPA background primary path	TRMMAP.HUBKGPRIPATH	

Flow Control

When **GTRANSPARA.TRANSCFGMODE** is set to **OLD**:

Table 4-25 Data to prepare for setting the flow control algorithm on the NodeB side

Parameter Name	Parameter ID	Setting Notes
Traffic Control Switch	RSCGRPALG.TCSW	See the recommended parameter configurations in 4.1.4 Application Scenarios .
Congestion Ctrl Switch	ULFLOWCTRLPARA.TNLCONGCTRLSWITCH	
Flow Control Switch	DLFLOWCTRLPARA.SWITCH	
Fair Switch	DLFLOWCTRLPARA.FAIRSWITCH	

When **GTRANSPARA.TRANSCFGMODE** is set to **NEW**:

Table 4-26 Data to prepare for setting the flow control algorithm on the NodeB side

Parameter Name	Parameter ID	Setting Notes
Traffic Control Switch	IPRSCGRPALG.TCSW	See the recommended parameter configurations in 4.1.4 Application Scenarios .
Congestion Ctrl Switch	IPULFLOWCTRLPARA.TNLCONGCTRLSWITCH	
Flow Control Switch	IPDLFLOWCTRLPARA.SWITCH	
Fair Flow Control Switch	IPDLFLOWCTRLPARA.FAIRSWITCH	

Other Data

Table 4-27 lists other data to prepare if access bandwidth is limited for multimode base stations.

Table 4-27 Other data to prepare if access bandwidth is limited for multimode base stations

Data Item	Sample Value	Remarks
Limited access bandwidth for a base station	20 Mbit/s	This item specifies the uplink and downlink limited access bandwidth for a base station.
Downlink bandwidth on the logical port of the RNC	20 Mbit/s	This item specifies the downlink limited access bandwidth for a base station.
Downlink bandwidth on the logical port of the BSC	10 Mbit/s	Calculates the bandwidth for this port based on the GSM traffic model of the base station.
GBTS index	1	-
Logical IP address of the GBTS	16.16.90.201	-
Abis IP address of the GBSC	172.16.140.140	-
Logical IP address of the NodeB	16.16.70.201	-
Iub IP address on the RNC side	172.16.100.140	-

Table 4-28 Other data to prepare if access bandwidth is limited for each operator in RAN sharing scenarios

Data Item	Sample Value	Remarks
Limited access bandwidth for operator A	10 Mbit/s	This item specifies the uplink and downlink limited access bandwidth for operator A.
Limited access bandwidth for operator B	10 Mbit/s	This item specifies the uplink and downlink limited access bandwidth for operator B.
Logical IP address of the NodeB (for operator A)	16.16.70.201	-
Logical IP address of the NodeB (for operator B)	16.16.60.201	-

Data Item	Sample Value	Remarks
Logical IP address of the eNodeB (for operator A)	16.15.70.201	-
Logical IP address of the eNodeB (for operator B)	16.15.60.201	-
Logical IP address of an lub port on the RNC side (for operator A)	172.16.90.140	-
Logical IP address of an lub port on the RNC side (for operator B)	172.16.80.140	-
Logical IP address of the serving gateway (S-GW) (for operator A)	172.15.90.140	-
Logical IP address of the serving gateway (S-GW) (for operator B)	172.15.80.140	-

Table 4-29 Other data to prepare if satellite transmission is used for GU dual-mode base station

Data Item	Sample Value	Remarks
Limited access bandwidth for a base station	4 Mbit/s	This item specifies the uplink and downlink limited access bandwidth for a base station.
Limited access bandwidth in UMTS mode	3 Mbit/s	This item specifies the downlink limited access bandwidth planned for UMTS, which must be lower than the downlink limited access bandwidth for the GU dual-mode base station.

Data Item	Sample Value	Remarks
Limited access bandwidth for GSM	3 Mbit/s	This item specifies the downlink limited access bandwidth planned for GSM, which must be lower than the downlink limited access bandwidth for the GU dual-mode base station.
GBTS index	1	-
Logical IP address of the GBTS	16.16.90.201	-
Abis IP address of the GBSC	172.16.140.140	-
Logical IP address of the NodeB	16.16.70.201	-
Iub IP address on the RNC side	172.16.100.140	-

4.4.2.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

4.4.2.2.1 Unlimited Access Bandwidth for Multimode Base Stations

Applicable Multimode Base Stations

- GULN/GUTN/GULTN
- GLN/GTN/GLTN
- ULN/UTN/ULTN
- LT/LN/TN/LTN
- GUL/GUT/GUN/GULT
- UL/UT/UN/ULT
- GL/GT/GN/GLT
- GU

The configuration process is similar in the multi-mode base station scenario. This section describes the process of configuring transmission resource management

strategies for GULN/GUTN/GULTN multimode base stations in unlimited access bandwidth scenarios.

Activation Command Examples (GSM Side)

For details about how to configure a TRM table on the BSC side, see the recommended configurations in [4.1.4.1.2 Transmission Resource Management Policy Configuration](#).

```
//Configuring a TRM table on the BSC side
//Setting the TRM mapping on the control plane on the Abis interface (BSC6900/BSC6910)
ADD TRMMAP:
TMI=111,ITFT=ABIS,TRANST=IP,CSVOICEPATH=EF,CSDATAPATH=AF41,PSHPRIDATAPATH=AF41,PSLPRIDATAPRI=AF31;
//Setting the TRM mapping on the control plane on the Abis interface (BSC6960)
ADD TRMMAP: TMI=111,ITFT=ABIS, CSVOICEPATH=EF, CSDATAPATH=AF41, PSHPRIDATAPATH=AF41, PSLPRIDATAPRI=AF31;

//Setting the TRM mapping on the control plane on the Abis interface
SET BSCABISPRIMAP: IDTYPE=BYID, BTSID=1, TRANSTYPE=IP, OMLDSCP=48, RSLDSCP=48, EMLDSCP=18, ESLDSCP=48;
//Adding the mapping from the Abis interface to the TRMMAP index (BSC6900/BSC6910)
ADD ADJMAP: ANI=3, ITFT=ABIS, TMIGLD=111, FTI=1;
//Adding the mapping from the Abis interface to the TRMMAP index (BSC6960)
ADD ADJMAP: ANI=3, TMIGLD=111;
//Configuring a TRM table on the base station side
//Setting the mapping between control- and user-plane data and DSCP values for the GBTS
SET BTSVLAN: IDTYPE=BYID, BTSID=1, SERVICETYPE=OML, DSCP=48;
SET BTSVLAN: IDTYPE=BYID, BTSID=1, SERVICETYPE=RSL, DSCP=48;
SET BTSVLAN: IDTYPE=BYID, BTSID=1, SERVICETYPE=EML, DSCP=18;
SET BTSVLAN: IDTYPE=BYID, BTSID=1, SERVICETYPE=ESL, DSCP=48;
SET BTSVLAN: IDTYPE=BYID, BTSID=1, SERVICETYPE= CSVOICE, DSCP=46;
SET BTSVLAN: IDTYPE=BYID, BTSID=1, SERVICETYPE= CSDATA, DSCP=34;
SET BTSVLAN: IDTYPE=BYID, BTSID=1, SERVICETYPE= PSHIGHPRI, DSCP=34;
SET BTSVLAN: IDTYPE=BYID, BTSID=1, SERVICETYPE= PSLOWPRI, DSCP=26;
SET BTSVLAN: IDTYPE=BYID, BTSID=1, SERVICETYPE= OTHERDATA, DSCP=46;
//Setting the mapping between control-plane data and DSCP values for the eGBTS
SET DIFPRI: PRIRULE=DSCP, SIGPRI=48, OMHIGHPRI=46, OMLOWPRI=18, IPCLKPRI=46;
//Optional Setting the separate mapping between control-plane data and DSCP values for the eGBTS
MOD SCTPLNK: SCTPNO=0, DSCPSW=ON, DSCP=48;
```

Activation Command Examples (UMTS Side)

For details about how to configure a TRM table on the RNC side, see the recommended configurations in [4.1.4.1.2 Transmission Resource Management Policy Configuration](#).

```
//Configuring a TRM table on the RNC side
//Setting the TRM mapping on the control and user planes on the lub interface (BSC6900/BSC6910)
ADD TRMMAP: TMI=110, ITFT=IUB, TRANST=IP, CCHPRIPATH=EF, SIPPRIPATH=EF, SRBPRIPATH=EF, VOICEPRIPATH=EF, CSCONVPRI=AF41, CSSTRMPRI=AF41, PSCONVPRI=AF41, PSSTRMPRI=AF41, PSINTHGHPRIPATH=AF21, PSINTLOWPRIPATH=AF21, PSBKGPRIPATH=AF21, HDSRBPRIPATH=EF, HDSIPPRIPATH=EF, HDVOICEPRIPATH=EF, HDCONVPRI=AF41, HDSTRMPRI=AF41, HDINTHGHPRIPATH=AF11, HDINTMIDPRIPATH=AF11, HDINTLOWPRIPATH=AF11, HDBKGPRIPATH=AF11, HUSRBPRIPATH=EF, HUSIPPRIPATH=EF, HUVOICEPRIPATH=EF, HUCONVPRI=AF41, HUSTRMPRI=AF41, HUINTHGHPRIPATH=AF11, HUINTMIDPRIPATH=AF11, HUINTLOWPRIPATH=AF11, HUBKGPRIPATH=AF11;
//Setting the TRM mapping on the control and user planes on the lub interface (BSC6960)
ADD TRMMAP: TMI=110, ITFT=IUB;

//Adding the mapping from the lub interface to the TRMMAP index (BSC6900/BSC6910)
ADD ADJMAP: ANI=10, ITFT=IUB, TRANST=IP, CNMNGMODE=SHARE, TMIGLD=110, TMISLV=110, TMIBRZ=110, FTI=1;
//Adding the mapping from the lub interface to the TRMMAP index (BSC6960)
ADD ADJMAP: ANI=10, TMIGLD=110, TMISLV=110, TMIBRZ=110;
//Configuring a TRM table on the NodeB side
```

```

SET DIFPRI: PRIRULE=DSCP, SIGPRI=48, OMHIGHPRI=46, OMLOWPRI=18, IPCLKPRI=46;
//(Optional) Setting the separate mapping between control-plane data and DSCP values for the NodeB in
link configuration mode
MOD SCTPLNK: SCTPNO=0, DSCPSW=ON, DSCP=48;
//(Optional) Setting the separate mapping between control-plane data and DSCP values for the NodeB in
endpoint configuration mode
MOD SCTPTemplate: SCTPTemplateID=0, DSCPSW=ON, DSCP=48;

//Configuring the dynamic flow control algorithm for the NodeB (when the bearer network supports two or
more queues)
//When GTRANSPARA.TRANSFCGMODE is set to OLD:
//Adding HSUPA flow control parameters
ADD ULFLOWCTRLPARA: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, BEAR=IP, PT=ETH, PN=0,
BWPRTSWITCH=ON, TNLCONGCTRLSWITCH=ON;
//Adding HSDPA flow control parameters
ADD DLFLOWCTRLPARA: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, BEAR=IP, PT=ETH, PN=0,
SWITCH=BW_SHAPING_ONOFF_TOGGLE, FAIRSWITCH=ON;

//When GTRANSPARA.TRANSFCGMODE is set to NEW:
//Adding HSUPA flow control parameters
ADD IPULFLOWCTRLPARA: IPULFLOWCTRLALGID=0, PT=ETH, PORTID=0, BWPRTSWITCH=ON,
TNLCONGCTRLSWITCH=ON;
//Adding HSDPA flow control parameters
ADD IPDLFLOWCTRLPARA: IPDLFLOWCTRLALGID=0, PT=ETH, PORTID=0,
SWITCH=BW_SHAPING_ONOFF_TOGGLE, FAIRSWITCH=ON;

```

Activation Command Examples (LTE Side)

For details about how to configure a TRM table on the eNodeB side, see the recommended configurations in [4.1.4.1.2 Transmission Resource Management Policy Configuration](#).

```

//Configuring a TRM table on the eNodeB side
//Setting the mapping between control-plane data and DSCP values for the eNodeB
SET DIFPRI: PRIRULE=DSCP, SIGPRI=48, OMHIGHPRI=46, OMLOWPRI=18, IPCLKPRI=46;
//(Optional) Setting the separate mapping between control-plane data and DSCP values for the eNodeB in
link configuration mode
MOD SCTPLNK: SCTPNO=0, DSCPSW=ON, DSCP=48;
//(Optional) Setting the separate mapping between control-plane data and DSCP values for the eNodeB in
endpoint configuration mode
MOD SCTPTemplate: SCTPTemplateID=0, DSCPSW=ON, DSCP=48;

//Setting the mapping between user-plane data and DSCP values for the eNodeB. When the control plane
ClO T EPS optimization function is implemented between the NB-IoT eNodeB and core network, no user-
plane data configuration is required. The configurations are as follows:
MOD UDTParAGR: UDTParAGRPID=40, PRIRULE=DSCP, PRI=46, ACTFACTOR=100;
MOD UDTParAGR: UDTParAGRPID=41, PRIRULE=DSCP, PRI=26, ACTFACTOR=100;
MOD UDTParAGR: UDTParAGRPID=42, PRIRULE=DSCP, PRI=34, ACTFACTOR=100;
MOD UDTParAGR: UDTParAGRPID=43, PRIRULE=DSCP, PRI=26, ACTFACTOR=100;
MOD UDTParAGR: UDTParAGRPID=44, PRIRULE=DSCP, PRI=46;
MOD UDTParAGR: UDTParAGRPID=45, PRIRULE=DSCP, PRI=18;
MOD UDTParAGR: UDTParAGRPID=46, PRIRULE=DSCP, PRI=18;
MOD UDTParAGR: UDTParAGRPID=47, PRIRULE=DSCP, PRI=18;
MOD UDTParAGR: UDTParAGRPID=48, PRIRULE=DSCP, PRI=0;

```

Activation Command Examples (NR Side)

For details about how to configure a TRM table on the NR side, see the recommended configurations in [4.1.4.1.2 Transmission Resource Management Policy Configuration](#).

```

//Configuring a TRM table on the NR side
//Setting the mapping between control-plane data and DSCP values for the NR side
SET DIFPRI: PRIRULE=DSCP, SIGPRI=48, OMHIGHPRI=46, OMLOWPRI=18, IPCLKPRI=46;
//(Optional) Setting the separate mapping between control-plane data and DSCP values for the NR side
MOD SCTPTemplate: SCTPTemplateID=0, DSCPSW=ON, DSCP=48;

```

```
//Setting the mapping between user-plane data and DSCP values for the NR side
MOD UDTPARAGRP: UDTPARAGRPID=40, PRIRULE=DSCP, PRI=46, ACTFACTOR=100;
MOD UDTPARAGRP: UDTPARAGRPID=41, PRIRULE=DSCP, PRI=26, ACTFACTOR=100;
MOD UDTPARAGRP: UDTPARAGRPID=42, PRIRULE=DSCP, PRI=34, ACTFACTOR=100;
MOD UDTPARAGRP: UDTPARAGRPID=43, PRIRULE=DSCP, PRI=26, ACTFACTOR=100;
MOD UDTPARAGRP: UDTPARAGRPID=44, PRIRULE=DSCP, PRI=46;
MOD UDTPARAGRP: UDTPARAGRPID=45, PRIRULE=DSCP, PRI=18;
MOD UDTPARAGRP: UDTPARAGRPID=46, PRIRULE=DSCP, PRI=18;
MOD UDTPARAGRP: UDTPARAGRPID=47, PRIRULE=DSCP, PRI=18;
MOD UDTPARAGRP: UDTPARAGRPID=48, PRIRULE=DSCP, PRI=0;
```

4.4.2.2.2 Limited Access Bandwidth for Multimode Base Stations

Applicable Multimode Base Stations

- GULN/GUTN/GULTN
- GLN/GTN/GLTN
- ULN/UTN/ULTN
- LN/TN/LTN
- GUL/GUT/GULT
- UL/UT/ULT
- GL/GT/GLT
- GU

The configuration process is similar in the multi-mode base station scenario. This section describes the process of configuring transmission resource management strategies for GULN/GUTN/GULTN multimode base stations in limited access bandwidth scenarios.

Activation Command Examples (GSM Side)

For details about how to configure a TRM table on the BSC side, see the recommended configuration in [4.1.4.1.2 Transmission Resource Management Policy Configuration](#).

```
//Configuring logical ports on the BSC side (Only the BSC6900 and BSC6910 are supported. The BSC6960 is not supported.)
//Adding a logical port on the Abis interface (BSC6900). In the following script, the bandwidth of this port is the CIR value (157) multiplied by 64, that is, 10048 kbit/s.
ADD IPLOGICPORT: SRN=1, SN=24, BT=GOUC, LPNTYPE=Leaf, LPN=1, CARRY=ETHER, PN=0,
RSCMNGMODE=SHARE, BWADJ=OFF, CIR=157, FLOWCTRLSWITCH=ON, OPSEPFLAG=OFF;
//Adding a logical port on the Abis interface (BSC6910). In the following script, the bandwidth of this port is the CIR value (157) multiplied by 64, that is, 10048 kbit/s.
ADD IPLOGICPORT: SRN=1, SN=24, BT=GOUC, LPNTYPE=Leaf, FLOWCTRLSWITCH=ON, CIR=157, LPN=1,
CARRY=IPPOOL, IPADDR="172.16.140.140";
//For a GBTS, binding the logical port on the Abis interface to the GBTS. In the following script, the base station is identified by its base station ID.
SET BTSIP: IDTYPE=BYID, BTSID=1, BTSCOMTYPE=LOGICIP, BTSIP="16.16.90.201", BSCIP="172.16.140.140",
CFGFLAG=IPLGCPOR, SN=24, LPN=1;
//For an eGBTS, binding an IP path to the logical port on the Abis interface if the peer end is a BSC6900
ADD IPPATH: ANI=3, PATHID=0, ITFT=ABIS, ISEGBTS=Yes, PATHT=QoS, IPADDR="172.16.140.140",
PEERIPADDR="16.16.90.201", TXBW=10000, RXBW=10000, CARRYFLAG=IPLGCPOR, LPNSN=0, LPN=1,
VLANFLAG=DISABLE, PATHCHK=DISABLED, AbisLnkBKFLAG=OFF;
//For an eGBTS, binding an adjacent node to the logical port on the Abis interface if the peer end is a BSC6910
ADD ADJLOGICPORTBIND: ANI=3, SRN=1, SN=24, LPN=1;

//Configuring a TRM table on the BSC side (Only the BSC6900 and BSC6910 are supported. The BSC6960 is not supported.)
```

```
//Setting the TRM mapping on the user plane on the Abis interface
ADD
TRMMAP:TMI=111,ITFT=ABIS,TRANST=IP,CSVOICEPATH=EF,CSDATAPATH=AF41,PSHPRIDATAPATH=AF41,PSL
PRIDATAPRI=AF31;
//Setting the TRM mapping on the control plane on the Abis interface
SET BSCABISPRIMAP: IDTYPE=BYID, BTSID=1, TRANSTYPE=IP, OMLDSCP=48, RSLDSCP=48, EMLDSCP=18,
ESLDSCP=48;
//Adding the mapping from the Abis interface to the TRMMAP index
ADD ADJMAP: ANI=3, ITFT=ABIS, TMIGLD=111, FTI=1;

//Configuring a TRM table on the base station side
//Setting the mapping between control- and user-plane data and DSCP values for the GBTS
SET BTSVLAN: IDTYPE=BYID, BTSID=1, SERVICETYPE=OML, DSCP=48;
SET BTSVLAN: IDTYPE=BYID, BTSID=1, SERVICETYPE=RSL, DSCP=48;
SET BTSVLAN: IDTYPE=BYID, BTSID=1, SERVICETYPE=EML, DSCP=18;
SET BTSVLAN: IDTYPE=BYID, BTSID=1, SERVICETYPE=ESL, DSCP=48;
SET BTSVLAN: IDTYPE=BYID, BTSID=1, SERVICETYPE= CSVOICE, DSCP=46;
SET BTSVLAN: IDTYPE=BYID, BTSID=1, SERVICETYPE= CSDATA, DSCP=34;
SET BTSVLAN: IDTYPE=BYID, BTSID=1, SERVICETYPE= PSHIGHPRI, DSCP=34;
SET BTSVLAN: IDTYPE=BYID, BTSID=1, SERVICETYPE= PSLOWPRI, DSCP=26;
SET BTSVLAN: IDTYPE=BYID, BTSID=1, SERVICETYPE= OTHERDATA, DSCP=46;
//Setting the mapping between control-plane data and DSCP values for the eGBTS
SET DIFPRI: PRIRULE=DSCP, SIGPRI=48, OMHIGHPRI=46, OMLOWPRI=18, IPCLKPRI=46;
//Optional) Setting the separate mapping between control-plane data and DSCP values for the eGBTS
MOD SCTPLNK: SCTPNO=0, DSCPSW=ON, DSCP=48;
```

Activation Command Examples (UMTS Side)

For details about how to configure a TRM table on the RNC side, see the recommended configuration in [4.1.4.1.2 Transmission Resource Management Policy Configuration](#).

```
//Configuring traffic limiting and shaping on the NodeB side
//When GTRANSPARA.TRANSFCFGMODE is set to OLD:
//Configuring traffic limiting and shaping if the NodeB side of a separate-MPT multimode base station
provides a co-transmission port
SET LR: CN=0, SRN=0, SN=6, SBT=BASE_BOARD, PT=ETH, PN=0, LRSW=ENABLE, CIR=20000, CBS=40000,
EBS=0;
//When GTRANSPARA.TRANSFCFGMODE is set to NEW:
//Configuring traffic limiting and shaping if the NodeB side of a separate-MPT multimode base station
provides a co-transmission port
ADD PORTLR: PORTLRID=0, PT=ETH, PORTID=0, CIR=20000, CBS=40000, EBS=0, DLCIR=2000;

//Configuring logical ports on the RNC side (Only the BSC6900 and BSC6910 are supported. The BSC6960 is
not supported.)
//Adding a logical port on the lub interface. In the following script, the bandwidth of this port is the CIR
value (313) multiplied by 64, that is, 20032 kbit/s.
ADD IPLOGICPORT: SRN=1, SN=26, BT=GOUC, LPNTYPE=Leaf, LPN=1, CARRYT=ETHER, PN=0,
RSCMNGMODE=SHARE, BWADJ=OFF, CIR=313, FLOWCTRLSWITCH=ON, OPSEFLAG=OFF;
//Binding an IP path to the logical port on the lub interface if the non-transmission-resource-pool
networking is used and the base station controller is a BSC6900
ADD IPPATH: ANI=10, PATHID=1, ITFT=IUB, TRANST=IP, PATHT=QoS, IPADDR="172.16.100.140",
PEERIPADDR="16.16.70.201", TXBW=20000, RXBW=20000, CARRYFLAG=NULL, VLANFLAG=DISABLE,
PATHCHK=DISABLED;
//Binding an adjacent node to the logical port on the lub interface if the transmission resource pool
networking is used
ADD ADJLOGICPORTBIND: ANI=10, SRN=1, SN=26, LPN=1;

//Configuring a TRM table on the RNC side (Only the BSC6900 and BSC6910 are supported. The BSC6960 is
not supported.)
//Setting the TRM mapping on the control and user planes on the lub interface
ADD
TRMMAP:TMI=110,ITFT=IUB,TRANST=IP,CCHPRIPATH=EF,SIPPRIPATH=EF,SRBPRIPATH=EF,VOICEPRIPATH=EF,
CSCONVPRIIPATH=AF41,CSSTRMPRIIPATH=AF41,PSCONVPRIIPATH=AF41,PSSTRMPRIIPATH=AF41,PSINTHGHP
RIIPATH=AF21,PSINTLOWPRIIPATH=AF21,PSBKGPRIPATH=AF21,HDSRBPRIPATH=EF,HDSIPPRIPATH=EF,HDVOI
CEPRIPATH=EF,HDCONVPRIIPATH=AF41,HDSTRMPRIIPATH=AF41,HDINTHGHPRIIPATH=AF11,HDINTMIDPRIIPATH=AF11,
HDINTLOWPRIIPATH=AF11,HDBKGPRIPATH=AF11,HUSRBPRIPATH=EF,HUSIPPRIPATH=EF,HUVOICE
PRIPATH=EF,HUCONVPRIIPATH=AF41,HUSTRMPRIIPATH=AF41,HUINTHGHPRIIPATH=AF11,HUINTMIDPRIIPATH=AF11,
HUBKGPRIPATH=AF11;
```

```
//Binding an adjacent node to the logical port on the lub interface if the transmission resource pool
networking is used
ADD ADJLOGICPORTBIND: ANI=10, SRN=1, SN=26, LPN=1;
//Configuring a TRM table on the NodeB side
SET DIFPRI: PRIRULE=DSCP, SIGPRI=48, OMHIGHPRI=46, OMLOWPRI=18, IPCLKPRI=46;
//((Optional) Setting the separate mapping between control-plane data and DSCP values for the NodeB in
link configuration mode
MOD SCTPLNK: SCTPNO=0, DSCPSW=ON, DSCP=48;
//((Optional) Setting the separate mapping between control-plane data and DSCP values for the NodeB in
endpoint configuration mode
MOD SCTPTemplate: SCTPTemplateID=0, DSCPSW=ON, DSCP=48;

//Configuring the dynamic flow control algorithm for the NodeB (when the bearer network supports two or
more queues)
//When GTRANSPARA.TRANSCFGMODE is set to OLD:
//Adding HSUPA flow control parameters
ADD ULFLOWCTRLPARA: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, BEAR=IP, PT=ETH, PN=0,
BWPRTSWITCH=ON, TNLCONGCTRLSWITCH=ON;
//Adding HSDPA flow control parameters
ADD DLFLOWCTRLPARA: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, BEAR=IP, PT=ETH, PN=0,
SWITCH=BW_SHAPING_ONOFF_TOGGLE, FAIRSWITCH=ON;

//When GTRANSPARA.TRANSCFGMODE is set to NEW:
//Adding HSUPA flow control parameters
ADD IPULFLOWCTRLPARA: IPULFLOWCTRLALGID=0, PT=ETH, PORTID=0, BWPRTSWITCH=ON,
TNLCONGCTRLSWITCH=ON;
//Adding HSDPA flow control parameters
ADD IPDLFLOWCTRLPARA: IPDLFLOWCTRLALGID=0, PT=ETH, PORTID=0,
SWITCH=BW_SHAPING_ONOFF_TOGGLE, FAIRSWITCH=ON;

//Turning off the traffic control switch of the default transmission resource group configured on the co-
transmission port if co-transmission is implemented between NR and GUL
//When GTRANSPARA.TRANSCFGMODE is set to OLD:
//Configuring a default transmission resource group on the co-transmission port
ADD RSCGRP: CN=0, SRN=0, SN=6, BEAR=IP, SBT=BASE_BOARD, PT=ETH, RSCGRPID=DEFAULTPORT,
RU=KBPS;
//Turning off the traffic control switch on the default transmission resource group you have configured
SET RSCGRPALG: CN=0, SRN=0, SN=6, BEAR=IP, SBT=BASE_BOARD, PT=ETH, RSCGRPID=DEFAULTPORT,
TCSW=DISABLE;

//When GTRANSPARA.TRANSCFGMODE is set to NEW:
//Configuring a default transmission resource group on the co-transmission port
ADD IPRSCGRP: IPRSCGRPID=0, PT=ETH, PORTID=0, RSCGRPNO=DEFAULTPORT, RU=KBPS,
TXBW=100000000, RXBW=100000000, TXCIR=100000000, RXCIR=100000000, TXPIR=100000000,
RXPIR=100000000, TXPBS=100000000;
//Turning off the traffic control switch on the default transmission resource group you have configured
SET IPRSCGRPALG: IPRSCGRPID=0, TCSW=DISABLE;
```

Activation Command Examples (LTE Side)

For details about how to configure a TRM table on the base station side, see the recommended configuration in [4.1.4.1.2 Transmission Resource Management Policy Configuration](#).

```
//Configuring traffic limiting and shaping on the base station side
//When GTRANSPARA.TRANSCFGMODE is set to OLD:
//Configuring traffic limiting and shaping if the eNodeB side of a separate-MPT multimode base station
provides a co-transmission port
SET LR: CN=0, SRN=0, SN=6, SBT=BASE_BOARD, PT=ETH, PN=0, LRSW=ENABLE, CIR=20000, CBS=40000,
EBS=0;

//When GTRANSPARA.TRANSCFGMODE is set to NEW:
//Configuring traffic limiting and shaping if the eNodeB side of a separate-MPT multimode base station
provides a co-transmission port
ADD PORTLR: PORTLRID=0, PT=ETH, PORTID=0, CIR=20000, CBS=40000, EBS=0, DLCIR=2000;

//Configuring a TRM table on the base station side
//Setting the mapping between control-plane data and DSCP values for the eNodeB
```

```

SET DIFPRI: PRIRULE=DSCP, SIGPRI=48, OMHIGHPRI=46, OMLWPRI=18, IPCLKPRI=46;
//(Optional) Setting the separate mapping between control-plane data and DSCP values for the eNodeB in
link configuration mode
MOD SCTPLNK: SCTPNO=0, DSCPSW=ON, DSCP=48;
//(Optional) Setting the separate mapping between control-plane data and DSCP values for the eNodeB in
endpoint configuration mode
MOD SCTPTemplate: SCTPTemplateID=0, DSCPSW=ON, DSCP=48;

//Setting the mapping between user-plane data and DSCP values for the eNodeB. When the control plane
ClOT EPS optimization function is implemented between the NB-IoT eNodeB and core network, no user-
plane data configuration is required. That is, the following configurations are not required.
MOD UDTParagrP: UDTParagrPID=40, PRIRULE=DSCP, PRI=46, ACTFACTOR=100;
MOD UDTParagrP: UDTParagrPID=41, PRIRULE=DSCP, PRI=26, ACTFACTOR=100;
MOD UDTParagrP: UDTParagrPID=42, PRIRULE=DSCP, PRI=34, ACTFACTOR=100;
MOD UDTParagrP: UDTParagrPID=43, PRIRULE=DSCP, PRI=26, ACTFACTOR=100;
MOD UDTParagrP: UDTParagrPID=44, PRIRULE=DSCP, PRI=46;
MOD UDTParagrP: UDTParagrPID=45, PRIRULE=DSCP, PRI=18;
MOD UDTParagrP: UDTParagrPID=46, PRIRULE=DSCP, PRI=18;
MOD UDTParagrP: UDTParagrPID=47, PRIRULE=DSCP, PRI=18;
MOD UDTParagrP: UDTParagrPID=48, PRIRULE=DSCP, PRI=0;

//Turning off the traffic control switch of the default transmission resource group configured on the co-
transmission port if co-transmission is implemented between NR and GUL
//When GTRANSPARA.TRANSFCFGMODE is set to OLD:
//Configuring a default transmission resource group on the co-transmission port
ADD RSCGRP: CN=0, SRN=0, SN=6, BEAR=IP, SBT=BASE_BOARD, PT=ETH, RSCGRPID=DEFAULTPORT,
RU=KBPS;
//Turning off the traffic control switch on the default transmission resource group you have configured
SET RSCGRPALG: CN=0, SRN=0, SN=6, BEAR=IP, SBT=BASE_BOARD, PT=ETH, RSCGRPID=DEFAULTPORT,
TCSW=DISABLE;
//When GTRANSPARA.TRANSFCFGMODE is set to NEW:
//Configuring a default transmission resource group on the co-transmission port
ADD IPRSCGRP: IPRSCGRPID=0, PT=ETH, PORTID=0, RSCGRPNO=DEFAULTPORT, RU=KBPS,
TXBW=100000000, RXBW=100000000, TXCIR=100000000, RXCIR=100000000, TXPIR=100000000,
RXPIR=100000000, TXPBS=100000000;
//Turning off the traffic control switch on the default transmission resource group you have configured
SET IPRSCGRPALG: IPRSCGRPID=0, TCSW=DISABLE;

```

When the user plane ClOT EPS optimization function is implemented on the NB-IoT base station and core network, NB-IoT does not support the traffic control switch for a transmission resource group.

Activation Command Examples (NR Side)

For details about how to configure a TRM table on the base station side, see the recommended configuration in [4.1.4.1.2 Transmission Resource Management Policy Configuration](#).

```

//Configuring traffic limiting and shaping on the base station side
//When GTRANSPARA.TRANSFCFGMODE is set to OLD:
//Configuring traffic limiting and shaping if the NR side of a separate-MPT multimode base station provides
a co-transmission port
SET LR: CN=0, SRN=0, SN=6, SBT=BASE_BOARD, PT=ETH, PN=0, LRSW=ENABLE, CIR=20000, CBS=40000,
EBS=0;
//When GTRANSPARA.TRANSFCFGMODE is set to NEW:
//Configuring traffic limiting and shaping if the NR side of a separate-MPT multimode base station provides
a co-transmission port
ADD PORTLR: PORTLRID=0, PT=ETH, PORTID=0, CIR=20000, CBS=40000, EBS=0, DLCIR=2000;

//Configuring a TRM table on the base station side
//Setting the mapping between control-plane data and DSCP values for the NR side
SET DIFPRI: PRIRULE=DSCP, SIGPRI=48, OMHIGHPRI=46, OMLWPRI=18, IPCLKPRI=46;
//(Optional) Setting the separate mapping between control-plane data and DSCP values for the NR side
MOD SCTPTemplate: SCTPTemplateID=0, DSCPSW=ON, DSCP=48;

//Setting the mapping between user-plane data and DSCP values for the NR side
MOD UDTParagrP: UDTParagrPID=40, PRIRULE=DSCP, PRI=46, ACTFACTOR=100;
MOD UDTParagrP: UDTParagrPID=41, PRIRULE=DSCP, PRI=26, ACTFACTOR=100;

```



```

MOD UDTPARAGRP: UDTPARAGRPID=42, PRIRULE=DSCP, PRI=34, ACTFACTOR=100;
MOD UDTPARAGRP: UDTPARAGRPID=43, PRIRULE=DSCP, PRI=26, ACTFACTOR=100;
MOD UDTPARAGRP: UDTPARAGRPID=44, PRIRULE=DSCP, PRI=46;
MOD UDTPARAGRP: UDTPARAGRPID=45, PRIRULE=DSCP, PRI=18;
MOD UDTPARAGRP: UDTPARAGRPID=46, PRIRULE=DSCP, PRI=18;
MOD UDTPARAGRP: UDTPARAGRPID=47, PRIRULE=DSCP, PRI=18;
MOD UDTPARAGRP: UDTPARAGRPID=48, PRIRULE=DSCP, PRI=0;

//Turning off the traffic control switch of the default transmission resource group configured on the co-
transmission port if co-transmission is implemented between NR and GUL
//When GTRANSPARA.TRANSFCFGMODE is set to OLD:
//Configuring a default transmission resource group on the co-transmission port
ADD RSCGRP: CN=0, SRN=0, SN=6, BEAR=IP, SBT=BASE_BOARD, PT=ETH, RSCGRPID=DEFAULTPORT,
RU=KBPS;
//Turning off the traffic control switch on the default transmission resource group you have configured
SET RSCGRPALG: CN=0, SRN=0, SN=6, BEAR=IP, SBT=BASE_BOARD, PT=ETH, RSCGRPID=DEFAULTPORT,
TCSW=DISABLE;
//When GTRANSPARA.TRANSFCFGMODE is set to NEW:
//Configuring a default transmission resource group on the co-transmission port
ADD IPRSCGRP: IPRSCGRPID=0, PT=ETH, PORTID=0, RSCGRPNO=DEFAULTPORT, RU=KBPS,
TXBW=100000000, RXBW=100000000, TXCIR=100000000, RXCIR=100000000, TXPIR=100000000,
RXPIR=100000000, TXPBS=100000000;
//Turning off the traffic control switch on the default transmission resource group you have configured
SET IPRSCGRPALG: IPRSCGRPID=0, TCSW=DISABLE;

```

4.4.2.2.3 Limited Access Bandwidth for Each Operator in a Multimode Base Station in RAN Sharing Scenarios

Applicable Multimode Base Stations

UL/UT

Activation Command Examples (UMTS Side)

Configuring traffic limiting and shaping if the NodeB side of a separate-MPT multimode base station provides a co-transmission port and the multimode base station is shared by two operators

- When **GTRANSPARA.TRANSFCFGMODE** is set to **OLD**:

```

//Configuring a transmission resource group
ADD RSCGRP: CN=0, SRN=0, SN=6, BEAR=IP, SBT=BASE_BOARD, PT=ETH, PN=0, RSCGRPID=1,
RU=KBPS, TXBW=10000, RXBW=10000, TXCBS=20000, TXEBS=64, OID=0, WEIGHT=100, TXCIR=10000,
RXCIR=10000, TXPIR=10000, RXPIR=10000, TXPBS=20000;
ADD RSCGRP: CN=0, SRN=0, SN=6, BEAR=IP, SBT=BASE_BOARD, PT=ETH, PN=0, RSCGRPID=2,
RU=KBPS, TXBW=10000, RXBW=10000, TXCBS=20000, TXEBS=64, OID=1, WEIGHT=100, TXCIR=10000,
RXCIR=10000, TXPIR=10000, RXPIR=10000, TXPBS=20000;

//Binding an IP path to the configured transmission resource group in link mode
ADD IPPATH: PATHID=1, TRANSFCFGMODE=OLD, SN=6, SBT=BASE_BOARD, PT=ETH, JNRSCGRP=
ENABLE, RSCGRPID=1, LOCALIP="16.16.70.201", PEERIP="172.16.90.140", PATHTYPE=ANY;
ADD NODEBPATH: PATHID=1;
ADD IPPATH: PATHID=2, TRANSFCFGMODE=OLD, SN=6, SBT=BASE_BOARD, PT=ETH, JNRSCGRP=
ENABLE, RSCGRPID=2, LOCALIP="16.16.70.201", PEERIP="172.16.90.140", PATHTYPE=ANY;
ADD NODEBPATH: PATHID=2;

//Binding an endpoint group to the configured transmission resource group in endpoint mode
ADD EPGROUP: EPGROUPID=0;
ADD EPGROUP: EPGROUPID=1;
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="16.16.70.201";
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="16.16.60.201";
ADD USERPLANEPEER: UPPEERID=0, IPVERSION=IPv4, PEERIPV4="172.16.90.140";
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv4, PEERIPV4="172.16.80.140";
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
ADD UPPEER2EPGRP: EPGROUPID=0, UPPEERID=0;

```

```

ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=1;
ADD EP2RSCGRP: CN=0, SRN=0, MT=ENDPOINT_GROUP, SN=6, ENDPOINTID=0, BEAR=IP,
SBT=BASE_BOARD, PT=ETH, RSCGRPID=1;
ADD EP2RSCGRP: CN=0, SRN=0, MT=ENDPOINT_GROUP, SN=6, ENDPOINTID=1, BEAR=IP,
SBT=BASE_BOARD, PT=ETH, RSCGRPID=2;

//Binding the passing-by data to the configured transmission resource group
ADD IP2RSCGRP: MAPIDX=0, SN=6, SBT=BASE_BOARD, PT=ETH, RSCGRPID=1, DSTIP="172.15.90.140",
DSTMASK="255.255.255.255";
ADD IP2RSCGRP: MAPIDX=0, SN=6, SBT=BASE_BOARD, PT=ETH, RSCGRPID=2, DSTIP="172.15.80.140",
DSTMASK="255.255.255.255";

```

- When **GTRANSPARA.TRANSFCGMODE** is set to **NEW**:

```

//Configuring a transmission resource group
ADD IPRSCGRP: IPRSCGRPID=1, PT=ETH, PORTID=0, RSCGRPNO=1, RU=KBPS, TXBW=10000,
RXBW=10000, TXCBS=20000, TXEBS=64, OID=0, WEIGHT=100, TXCIR=10000, RXCIR=10000,
TXPIR=10000, RXPIR=10000, TXPBS=20000;
ADD IPRSCGRP: IPRSCGRPID=2, PT=ETH, PORTID=0, RSCGRPNO=2, RU=KBPS, TXBW=10000,
RXBW=10000, TXCBS=20000, TXEBS=64, OID=1, WEIGHT=100, TXCIR=10000, RXCIR=10000,
TXPIR=10000, RXPIR=10000, TXPBS=20000;

//Binding an IP path to the configured transmission resource group in link mode
ADD IPPATH: PATHID=1, TRANSFCGMODE=NEW, JNIPRSCGRP=ENABLE, IPRSCGRPID=1, BPT= ETH,
PORTID=0, LOCALIP="16.16.70.201", PEERIP="172.16.80.140", PATHTYPE=ANY;
ADD NODEBPATH: PATHID=1;
ADD IPPATH: PATHID=2, TRANSFCGMODE=NEW, JNIPRSCGRP=ENABLE, IPRSCGRPID=2, BPT= ETH,
PORTID=0, LOCALIP="16.16.70.201", PEERIP="172.16.80.140", PATHTYPE=ANY;
ADD NODEBPATH: PATHID=2;

//Binding an endpoint group to the configured transmission resource group in endpoint mode
ADD EPGROUP: EPGROUPID=0;
ADD EPGROUP: EPGROUPID=1;
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="16.16.70.201";
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="16.16.60.201";
ADD USERPLANEPEER: UPPEERID=0, IPVERSION=IPv4, PEERIPV4="172.16.90.140";
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv4, PEERIPV4="172.16.80.140";
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
ADD UPPEER2EPGRP: EPGROUPID=0, UPPEERID=0;
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=1;
ADD EP2IPRSCGRP: MAPID=0, MT= ENDPOINT_GROUP, ENDPOINTID=0, IPRSCGRPID=1;
ADD EP2IPRSCGRP: MAPID=1, MT= ENDPOINT_GROUP, ENDPOINTID=1, IPRSCGRPID=2;

//Binding the passing-by data to the configured transmission resource group
ADD IP2IPRSCGRP: IP2IPRSCGRPID=0, IPRSCGRPID=1, IPVERSION= IPV4, DSTIPV4="172.15.90.140",
DSTMASK="255.255.255.255";
ADD IP2IPRSCGRP: IP2IPRSCGRPID=1, IPRSCGRPID=2, IPVERSION= IPV4, DSTIPV4="172.15.80.140",
DSTMASK="255.255.255.255";

```

On the RNC side (Only the BSC6900 and BSC6910 are supported. The BSC6960 is not supported.)

```

//Configuring logical ports on the RNC side
//Adding a logical port on the lub interface
ADD IPLOGICPORT: SRN=1, SN=26, BT=GOUC, LPNTYPE=Leaf, LPN=1, CARRYT=ETHER, PN=0,
RSCMNGMODE=EXCLUSIVE, BWADJ=OFF, CIR=157, FLOWCTRLSWITCH=ON, OPSEPFLAG=OFF;
ADD IPLOGICPORT: SRN=1, SN=26, BT=GOUC, LPNTYPE=Leaf, LPN=2, CARRYT=ETHER, PN=0,
RSCMNGMODE=EXCLUSIVE, BWADJ=OFF, CIR=157, FLOWCTRLSWITCH=ON, OPSEPFLAG=OFF;

//Binding an IP path to the logical port on the lub interface when the base station controller is a BSC6900
ADD IPPATH: ANI=10, PATHID=1, ITFT=IUB, TRANST=IP, PATHT=QoS, IPADDR="172.16.90.140",
PEERIPADDR="16.16.70.201", TXBW=10000, RXBW=10000, CARRYFLAG=NULL, VLANFLAG=DISABLE,
PATHCHK=DISABLED;
ADD IPPATH: ANI=10, PATHID=2, ITFT=IUB, TRANST=IP, PATHT=QoS, IPADDR="172.16.80.140",
PEERIPADDR="16.16.60.201", TXBW=10000, RXBW=10000, CARRYFLAG=NULL, VLANFLAG=DISABLE,
PATHCHK=DISABLED;

//Configuring a TRM table on the RNC side
//Setting the TRM mapping on the control and user planes on the lub interface
ADD

```



```
TRMMAP:TMI=110,ITFT=IUB,TRANST=IP,CCHPRIPATH=EF,SIPPRIPATH=EF,SRBPRIPATH=EF,VOICEPRIPATH=EF,
CSCONVPRIATH=AF41,CSSTRMPRIATH=AF41,PSCONVPRIATH=AF41,PSSTRMPRIATH=AF41,PSINTHGHPRIPATH=AF21,PSINTLOWPRIPATH=AF21,PSBKGPRIPATH=AF21,HDSRBPRIPATH=EF,HDSIPPRIPATH=EF,HDVOI
CEPRIPATH=EF,HDCONVPRIATH=AF41,HDSTRMPRIATH=AF41,HDINTHGHPRIPATH=AF11,HDINTMIDPRIPATH=AF11,HDINTLOWPRIPATH=AF11,HDBKGPRIPATH=AF11,HUSRBPRIPATH=EF,HUSIPPRIPATH=EF,HUVOICE
PRIPATH=EF,HUCONVPRIATH=AF41,HUSTRMPRIATH=AF41,HUINTHGHPRIPATH=AF11,HUINTMIDPRIPATH=AF11,HUINTLOWPRIPATH=AF11,HUBKGPRIPATH=AF11;
```

```
//Adding the mapping from the lub interface to the TRMMAP index
ADD ADJMAP: ANI=10, ITFT=IUB, TRANST=IP, CNMNGMODE=SHARE, TMIGLD=110, TMISLV=110,
TMIBRZ=110, FTI=1;
```

Configuring the NodeB

```
//Configuring a TRM table on the NodeB side
//Setting the mapping between control-plane data and DSCP values for the NodeB
SET DIFPRI: PRIRULE=DSCP, SIGPRI=48, OMHIGHPRI=46, OMLOWPRI=18, IPCLKPRI=46;
//((Optional) Setting the separate mapping between control-plane data and DSCP values for the NodeB in
link configuration mode
MOD SCTPLNK: SCTPNO=0, DSCPSW=ON, DSCP=48;
//((Optional) Setting the separate mapping between control-plane data and DSCP values for the NodeB in
endpoint configuration mode
MOD SCTPTemplate: SCTPTemplateID=0, DSCPSW=ON, DSCP=48;

//Configuring the dynamic flow control algorithm for the NodeB (when the bearer network supports two or
more queues)
//When GTRANSPARA.TRANSFCGMODE is set to OLD:
//Adding HSUPA flow control parameters
ADD ULFLOWCTRLPARA: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, BEAR=IP, PT=ETH, PN=0,
BWPRTSWITCH=ON, TNLCONGCTRLSWITCH=ON;
//Adding HSDPA flow control parameters
ADD DLFLOWCTRLPARA: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, BEAR=IP, PT=ETH, PN=0,
SWITCH=BW_SHAPING_ONOFF_TOGGLE, FAIRSWITCH=ON;
//When GTRANSPARA.TRANSFCGMODE is set to NEW:
//Adding HSUPA flow control parameters
ADD IPULFLOWCTRLPARA: IPULFLOWCTRLALGID=0, PT=ETH, PORTID=0, BWPRTSWITCH=ON,
TNLCONGCTRLSWITCH=ON;
//Adding HSDPA flow control parameters
ADD IPDLFLOWCTRLPARA: IPDLFLOWCTRLALGID=0, PT=ETH, PORTID=0,
SWITCH=BW_SHAPING_ONOFF_TOGGLE, FAIRSWITCH=ON;
```

If the NodeB side of a separate-MPT multimode base station provides a co-transmission port and co-transmission is implemented through panel interconnection, the traffic control switch for the transmission resource group must be turned off to prevent passing-by data from preempting transmission resources of local data in the case of congestion, which deteriorates user experience.

```
//When GTRANSPARA.TRANSFCGMODE is set to OLD:
//Configuring transmission resource groups on the co-transmission port
SET RSCGRPALG: CN=0, SRN=0, SN=6, BEAR=IP, SBT=BASE_BOARD, PT=ETH, RSCGRPID=1, TCSW=DISABLE;
SET RSCGRPALG: CN=0, SRN=0, SN=6, BEAR=IP, SBT=BASE_BOARD, PT=ETH, RSCGRPID=2, TCSW=DISABLE;
//When GTRANSPARA.TRANSFCGMODE is set to NEW:
//Configuring transmission resource groups on the co-transmission port
SET IPRSCGRPALG: IPRSCGRPID=1, TCSW=DISABLE;
SET IPRSCGRPALG: IPRSCGRPID=2, TCSW=DISABLE;
```

Activation Command Examples (LTE Side)

Configuring traffic limiting and shaping if the eNodeB side of a separate-MPT multimode base station provides a co-transmission port and the multimode base station is shared by two operators

- When **GTRANSPARA.TRANSFCGMODE** is set to **OLD**:
Only the new transmission configuration model is supported in IPv6 transmission.

//Configuring a transmission resource group

```
ADD RSCGRP: CN=0, SRN=0, SN=6, BEAR=IP, SBT=BASE_BOARD, PT=ETH, PN=0, RSCGRPID=1,
RU=KBPS, TXBW=10000, RXBW=10000, TXCBS=20000, TXEBS=64, OID=0, WEIGHT=100, TXCIR=10000,
RXCIR=10000, TXPIR=10000, RXPIR=10000, TXPBS=10000;
ADD RSCGRP: CN=0, SRN=0, SN=6, BEAR=IP, SBT=BASE_BOARD, PT=ETH, PN=0, RSCGRPID=2,
RU=KBPS, TXBW=10000, RXBW=10000, TXCBS=20000, TXEBS=64, OID=1, WEIGHT=100, TXCIR=10000,
RXCIR=10000, TXPIR=10000, RXPIR=10000, TXPBS=10000;
```

//Link mode: Binding an IP path to the configured transmission resource group. If the control plane CloT EPS optimization function is implemented between the NB-IoT eNodeB and EPC, no user-plane data configuration is required.

```
ADD IPPATH: PATHID=1, TRANSCFGMODE=OLD, SN=6, SBT=BASE_BOARD, PT=ETH,
JNRSCGRP=ENABLE, RSCGRPID=1, LOCALIP="16.15.70.201", PEERIP="172.15.90.140", PATHTYPE=ANY;
ADD ENODEBPATH: IpPathId=1, AppType=S1;
ADD IPPATH: PATHID=2, TRANSCFGMODE=OLD, SN=6, SBT=BASE_BOARD, PT=ETH,
JNRSCGRP=ENABLE, RSCGRPID=2, LOCALIP="16.15.60.201", PEERIP="172.15.80.140", PATHTYPE=ANY;
ADD ENODEBPATH: IpPathId=2, AppType=S1;
```

//Endpoint mode: Binding an endpoint group to the configured transmission resource group. If the control plane CloT EPS optimization function is implemented between the NB-IoT eNodeB and EPC, no user-plane data configuration is required.

```
ADD EPGROUP: EPGROUPID=0;
ADD EPGROUP: EPGROUPID=1;
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="16.15.70.201";
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="16.15.60.201";
ADD USERPLANEPEER: UPPEERID=0, IPVERSION=IPv4, PEERIPV4="172.15.90.140";
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv4, PEERIPV4="172.15.80.140";
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
ADD UPPEER2EPGRP: EPGROUPID=0, UPPEERID=0;
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=1;
ADD EP2RSCGRP: CN=0, SRN=0, MT=ENDPOINT_GROUP, SN=6, ENDPOINTID=0, BEAR=IP,
SBT=BASE_BOARD, PT=ETH, RSCGRPID=1;
ADD EP2RSCGRP: CN=0, SRN=0, MT=ENDPOINT_GROUP, SN=6, ENDPOINTID=1, BEAR=IP,
SBT=BASE_BOARD, PT=ETH, RSCGRPID=2;
```

//Binding the passing-by data to the configured transmission resource group

```
ADD IP2RSCGRP: MAPIDX=0, SN=6, SBT=BASE_BOARD, PT=ETH, RSCGRPID=1, DSTIP="172.16.90.140",
DSTMASK="255.255.255.255";
ADD IP2RSCGRP: MAPIDX=1, SN=6, SBT=BASE_BOARD, PT=ETH, RSCGRPID=2, DSTIP="172.16.80.140",
DSTMASK="255.255.255.255";
```

- When **GTRANSPARA.TRANSCFGMODE** is set to **NEW**:

//Configuring a transmission resource group

```
ADD IPRSCGRP: IPRSCGRPID=1, PT=ETH, PORTID=0, RSCGRPNO=1, RU=KBPS, TXBW=10000,
RXBW=10000, TXCBS=20000, TXEBS=64, OID=0, WEIGHT=100, TXCIR=10000, RXCIR=10000,
TXPIR=10000, RXPIR=10000, TXPBS=20000;
ADD IPRSCGRP: IPRSCGRPID=2, PT=ETH, PORTID=0, RSCGRPNO=2, RU=KBPS, TXBW=10000,
RXBW=10000, TXCBS=20000, TXEBS=64, OID=1, WEIGHT=100, TXCIR=10000, RXCIR=10000,
TXPIR=10000, RXPIR=10000, TXPBS=20000;
```

//Link mode: Binding an IP path to the configured transmission resource group. If the control plane CloT EPS optimization function is implemented between the NB-IoT eNodeB and EPC, no user-plane data configuration is required. IPv6 does not support the link mode.

```
ADD IPPATH: PATHID=1, TRANSCFGMODE=NEW, JNIPRSCGRP=ENABLE, IPRSCGRPID=1,
LOCALIP="16.16.70.201", PEERIP="172.16.80.140", PATHTYPE=ANY;
ADD ENODEBPATH: IpPathId=1, AppType=S1;
ADD IPPATH: PATHID=2, TRANSCFGMODE=NEW, JNIPRSCGRP=ENABLE, IPRSCGRPID=2,
LOCALIP="16.16.70.201", PEERIP="172.16.80.140", PATHTYPE=ANY;
ADD ENODEBPATH: IpPathId=2, AppType=S1;
```

//Endpoint mode: Binding an endpoint group to the configured transmission resource group. If the control plane CloT EPS optimization function is implemented between the NB-IoT eNodeB and EPC, no user-plane data configuration is required. (IPv4)

```
ADD EPGROUP: EPGROUPID=0;
ADD EPGROUP: EPGROUPID=1;
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="16.15.70.201";
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="16.15.60.201";
ADD USERPLANEPEER: UPPEERID=0, IPVERSION=IPv4, PEERIPV4="172.15.90.140";
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv4, PEERIPV4="172.15.80.140";
```

```

ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
ADD UPPEER2EPGRP: EPGROUPID=0, UPPEERID=0;
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=1;
ADD EP2IPRSCGRP: MAPID=0, MT= ENDPOINT_GROUP, ENDPOINTID=0, IPRSCGRPID=1;
ADD EP2IPRSCGRP: MAPID=1, MT= ENDPOINT_GROUP, ENDPOINTID=1, IPRSCGRPID=2;
//Endpoint mode: Binding an endpoint group to the configured transmission resource group. If the
control plane CloT EPS optimization function is implemented between the NB-IoT eNodeB and EPC,
no user-plane data configuration is required. (IPv6)
ADD EPGROUP: EPGROUPID=0;
ADD EPGROUP: EPGROUPID=1;
ADD USERPLANEHOST: UPHOSTID=3, IPVERSION=IPv6, LOCIPV6="2001:db8:100:ad1:200:100:100:0";
ADD USERPLANEHOST: UPHOSTID=3, IPVERSION=IPv6, LOCIPV6="2001:db8:100:ad1:200:100:100:0";
ADD USERPLANEPEER: UPPEERID=2, IPVERSION=IPv6, PEERIPV6="2001:db8:100:ad1:200:100:3001:2";
ADD USERPLANEPEER: UPPEERID=3, IPVERSION=IPv6, PEERIPV6="2001:db8:100:ad1:200:100:3001:0";
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=2;
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=3;
ADD UPPEER2EPGRP: EPGROUPID=0, UPPEERID=2;
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=3;

//Binding the passing-by data to the configured transmission resource group (IPv4)
ADD IP2IPRSCGRP: IP2IPRSCGRPID=0, IPRSCGRPID=1, IPVERSION= IPV4, DSTIPV4="172.15.90.140",
DSTMASK="255.255.255.255";
ADD IP2IPRSCGRP: IP2IPRSCGRPID=1, IPRSCGRPID=2, IPVERSION= IPV4, DSTIPV4="172.15.80.140",
DSTMASK="255.255.255.255";
//Binding the passing-by data to the configured transmission resource group (IPv6)
ADD IP2IPRSCGRP: IP2IPRSCGRPID=0, IPRSCGRPID=1, IPVERSION= IPV6, DSTIPV6="1000:1111::",
PFXLEN=16;
ADD IP2IPRSCGRP: IP2IPRSCGRPID=1, IPRSCGRPID=2, IPVERSION= IPV6, DSTIPV6="1000:1112::",
PFXLEN=16;

```

Configuring a TRM table on the base station side

```

//Setting the mapping between control-plane data and DSCP values for the eNodeB
SET DIFPRI: PRIRULE=DSCP, SIGPRI=48, OMHIGHPRI=46, OMLWPRI=18, IPCLKPRI=46;
//((Optional) Setting the separate mapping between control-plane data and DSCP values for the eNodeB in
link configuration mode
MOD SCTPLNK: SCTPNO=0, DSCPSW=ON, DSCP=48;
//((Optional) Setting the separate mapping between control-plane data and DSCP values for the eNodeB in
endpoint configuration mode
MOD SCTPTemplate: SCTPTemplateID=0, DSCPSW=ON, DSCP=48;

//Setting the mapping between DSCP values and user-plane data for the eNodeB
MOD UDTPARAGRP: UDTPARAGRPID=40, PRIRULE=DSCP, PRI=46, ACTFACTOR=100;
MOD UDTPARAGRP: UDTPARAGRPID=41, PRIRULE=DSCP, PRI=26, ACTFACTOR=100;
MOD UDTPARAGRP: UDTPARAGRPID=42, PRIRULE=DSCP, PRI=34, ACTFACTOR=100;
MOD UDTPARAGRP: UDTPARAGRPID=43, PRIRULE=DSCP, PRI=26, ACTFACTOR=100;
MOD UDTPARAGRP: UDTPARAGRPID=44, PRIRULE=DSCP, PRI=46;
MOD UDTPARAGRP: UDTPARAGRPID=45, PRIRULE=DSCP, PRI=18;
MOD UDTPARAGRP: UDTPARAGRPID=46, PRIRULE=DSCP, PRI=18;
MOD UDTPARAGRP: UDTPARAGRPID=47, PRIRULE=DSCP, PRI=18;
MOD UDTPARAGRP: UDTPARAGRPID=48, PRIRULE=DSCP, PRI=0;

```

If the eNodeB side of a separate-MPT multimode base station provides a co-transmission port and co-transmission is implemented through panel interconnection, the traffic control switch for the transmission resource group must be turned off to prevent passing-by data from preempting transmission resources of local data in the case of congestion, which deteriorates user experience. NB-IoT does not support the traffic control switch for a transmission resource group.

```

//When GTRANSPARA. TRANSCFGMODE is set to OLD:
//Configuring transmission resource groups on the co-transmission port
SET RSCGRPAlG: CN=0, SRN=0, SN=6, BEAR=IP, SBT=BASE_BOARD, PT=ETH, RSCGRPID=1, TCSW=DISABLE;
SET RSCGRPAlG: CN=0, SRN=0, SN=6, BEAR=IP, SBT=BASE_BOARD, PT=ETH, RSCGRPID=2, TCSW=DISABLE;
//When GTRANSPARA. TRANSCFGMODE is set to NEW:
//Configuring transmission resource groups on the co-transmission port
SET IPRSCGRPAlG: IPRSCGRPID=1, TCSW=DISABLE;
SET IPRSCGRPAlG: IPRSCGRPID=2, TCSW=DISABLE;

```

4.4.2.2.4 GU Dual-Mode Base Stations Using Satellite Transmission

Applicable Multimode Base Stations

GU

Activation Command Examples (GSM Side)

The command configurations are the same as those for the GSM side in multimode base stations with limited access bandwidth. For details, see [Activation Command Examples \(GSM Side\)](#).

Activation Command Examples (UMTS Side)

Configuring traffic limiting and shaping on the base station side

```
//When GTRANSPARA.TRANSCFGMODE is set to OLD:
//Configuring traffic limiting and shaping if the NodeB side of a separate-MPT multimode base station
provides a co-transmission port
SET LR: CN=0, SRN=0, SN=6, SBT=BASE_BOARD, PT=ETH, PN=0, LRSW=ENABLE, CIR=4000, CBS=8000,
EBS=0;
//When GTRANSPARA.TRANSCFGMODE is set to NEW:
//Configuring traffic limiting and shaping if the NodeB side of a separate-MPT multimode base station
provides a co-transmission port
ADD PORTLR: PORTLRID=0, PT=ETH, PORTID=0, CIR=20000, CBS=40000, EBS=0, DLCIR=2000;
```

Configuring logical ports on the RNC side

```
//Adding a logical port on the lub interface
ADD IPLOGICPORT: SRN=1, SN=26, BT=GOUc, LPNTYPE=Leaf, LPN=1, CARRYT=ETHER, PN=0,
RSCMNGMODE=SHARE, BWADJ=OFF, CIR=47, FLOWCTRLSWITCH=ON, OPSEPFLAG=OFF;
//In the preceding script, the unit of bandwidth configured on a logical port is 64 kbit/s. Therefore, the CIR
value 47 means that the configured bandwidth is 3008 kbit/s.
//Binding an IP path to the logical port on the lub interface if the non-transmission-resource-pool
networking is used and the base station controller is a BSC6900
ADD IPPATH: ANI=10, PATHID=1, ITFT=IUB, TRANST=IP, PATHT=QoS, IPADDR="172.16.100.140",
PEERIPADDR="16.16.70.201", TXBW=3000, RXBW=3000, CARRYFLAG=IPLGCPORT, LPNSN=0, LPN=1,
PATHCHK=DISABLED;
//Binding an adjacent node to the logical port on the lub interface if the transmission resource pool
networking is used
ADD ADJLOGICPORTBIND: ANI=10, SRN=1, SN=26, LPN=1;
//Binding an SCTP link to the logical port on the lub interface
ADD SCTPLNK: SCTPLNKID=12, APP=NBAP, MODE=SERVER, SpecifyLOCPNFlag=NO,
LOCIP1="172.16.100.140", PEERIP1="16.16.70.201", PEERPN=2010, LOGPORTFLAG=YES, LOGPORTSN=0,
LOGPORTNO=1, SWITCHBACKFLAG=YES;
```

For details about how to configure a TRM table on the RNC side, see the recommended configuration in [4.1.4.4.2 Transmission Resource Management Strategies](#).

```
//Setting the TRM mapping on the control and user planes on the lub interface (BSC6900/BSC6910)
ADD
TRMMAP:TMI=110,ITFT=IUB,TRANST=IP,CCHPRIPATH=EF,SIPPRIPATH=EF,SRBPRIPATH=EF,VOICEPRIPATH=EF,
CSCONVPRIIPATH=AF41,CSSTRMPRIIPATH=AF41,PSCONVPRIIPATH=AF41,PSSTRMPRIIPATH=AF41,PSINTHGHP
RIIPATH=AF21,PSINTLOWPRIIPATH=AF21,PSBKGPRIPATH=AF21,HDSRBPRIPATH=EF,HDSIPPRIPATH=EF,HDOI
CEPRIPATH=EF,HDCONVPRIIPATH=AF41,HDSTRMPRIIPATH=AF41,HDINTHGHPRIIPATH=AF11,HDINTMIDPRI
PATH=AF11,HDINTLOWPRIIPATH=AF11,HDBKGPRIPATH=AF11,HUSRBPRIPATH=EF,HUSIPPRIPATH=EF,HUVOI
CEPRIPATH=EF,HUCONVPRIIPATH=AF41,HUSTRMPRIIPATH=AF41,HUINTHGHPRIIPATH=AF11,HUINTMIDPRI
PATH=AF11,HUINTLOWPRIIPATH=AF11,HUBKGPRIPATH=AF11;
//Setting the TRM mapping on the control and user planes on the lub interface (BSC6960)
ADD TRMMAP: TMI=1, ITFT=IUB;
//Adding the mapping from the lub interface to the TRMMAP index (BSC6900/BSC6910)
ADD ADJMAP: ANI=10, ITFT=IUB, TRANST=IP, CNMNGMODE=SHARE, TMIGLD=110, TMISLV=110,
TMIBRZ=110, FTI=1;
//Adding the mapping from the lub interface to the TRMMAP index (BSC6960)
```

```
ADD ADJMAP: ANI=10, TMIGLD=110, TMISLV=110, TMIBRZ=110;
//Configuring a TRM table on the NodeB side
SET DIFPRI: PRIRULE=DSCP, SIGPRI=48, OMHIGHPRI=46, OMLOWPRI=18, IPCLKPRI=46;
//(Optional) Setting the separate mapping between control-plane data and DSCP values for the NodeB in
link configuration mode
MOD SCTPLNK: SCTPNO=0, DSCPSW=ON, DSCP=48;
//(Optional) Setting the separate mapping between control-plane data and DSCP values for the NodeB in
endpoint configuration mode
MOD SCTPTemplate: SCTPTemplateID=0, DSCPSW=ON, DSCP=48;
```

Configuring the uplink bandwidth adaptive flow control switch and HSDPA flow control switch on the NodeB side

```
//When GTRANSPARA.TRANSFCGMODE is set to OLD:
//Adding HSUPA flow control parameters
ADD ULFLOWCTRLPARA: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, BEAR=IP, PT=ETH, PN=0,
BWPRTSWITCH=ON, TNLCONGCTRLSWITCH=ON;
//Adding HSDPA flow control parameters
ADD DLFLOWCTRLPARA: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, BEAR=IP, PT=ETH, PN=0,
SWITCH=BW_SHAPING_ONOFF_TOGGLE;
//When GTRANSPARA.TRANSFCGMODE is set to NEW:
//Adding HSUPA flow control parameters
ADD IPULFLOWCTRLPARA: IPULFLOWCTRLALGID=0, PT=ETH, PORTID=0, BWPRTSWITCH=ON,
TNLCONGCTRLSWITCH=ON;
//Adding HSDPA flow control parameters
ADD IPDLFLOWCTRLPARA: IPDLFLOWCTRLALGID=0, PT=ETH, PORTID=0,
SWITCH=BW_SHAPING_ONOFF_TOGGLE, FAIRSWITCH=ON;
```

4.4.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.3 Activation Verification

4.4.3.1 Unlimited Access Bandwidth for Multimode Base Stations

After the Bandwidth Sharing of Multimode Base Station Co-Transmission feature is activated, check whether UEs can properly process CS and PS services when transmission resources are congested and whether the DSCP value of each packet is configured as expected.

- If yes to both, this feature has been activated.
- If no to either, this feature has not been activated.

Perform the following steps to determine whether this feature has been activated:

Step 1 Start IP or MAC tracing on the LMT.

- If the eGBTS provides a co-transmission port, start IP or MAC tracing on the eGBTS LMT.
For IP tracing: Choose **Trace > Common Services > IP Layer Protocol Trace**.
For MAC tracing: Choose **Trace > Common Services > MAC Trace**.
- If the NodeB provides a co-transmission port, start IP or MAC tracing on the NodeB LMT.
For IP tracing: Choose **Trace > Common Services > IP Layer Protocol Trace**.

For MAC tracing: Choose **Trace > Common Services > MAC Trace**.

- If the eNodeB provides a co-transmission port, start IP or MAC tracing on the eNodeB LMT.

For IP tracing: Choose **Trace > Common Services > IP Layer Protocol Trace**.

For MAC tracing: Choose **Trace > Common Services > MAC Trace**.

- If the gNodeB provides a co-transmission port, start IP or MAC tracing on the gNodeB LMT.

For IP tracing: Choose **Trace > Common Services > IP Layer Protocol Trace**.

For MAC tracing: Choose **Trace > Common Services > MAC Trace**.

Step 2 For IP tracing, in the displayed **IP Layer Protocol Trace** dialog box, specify **Local IP Address** and **Peer IP Address** of the packets to be traced. For MAC tracing, in the displayed **MAC Trace** dialog box, specify **Local MAC Address** and **Peer MAC Address** of the packets to be traced.

Step 3 Use the TrafficReview tool to check the TOS field in the layer 3 IP packet header or the VLAN Priority field in the layer 2 IP packet header. The first six bits in the TOS field indicate the DSCP value of a packet. If the calculated DSCP values or VLAN priorities are the same as the planned DSCP values or VLAN priorities, this feature has been activated.

----End

4.4.3.2 Limited Access Bandwidth for Multimode Base Stations

If you do not need to check whether the configured service priority has taken effect, perform the following steps to check whether the feature has been activated:

Step 1 Run the **LST RSCGRP** (in the old model)/**LST IPRSCGRP** (in the new model) command to check whether the transmission resource group has been configured for the co-transmission port. If not, this activation observation method is not applicable.

Step 2 Initiate a UMTS, LTE, or NR PS service and set the maximum data rate to a value greater than the **CIR** value to simulate transmission resource congestion.

Step 3 Query the value of the VS.RscGroup.TxFLOWOverloadTime counter for the co-transmission port. If the value is greater than 0, this feature has been activated.

----End

If you need to check whether the configured service priority has taken effect, perform the following steps to check whether the feature has been activated:

eGBTS of a Multimode Base Station Providing a Co-transmission Port

Step 1 Initiate a UMTS, LTE, or NR PS service and set the maximum data rate to a value greater than the **CIR** value to simulate transmission resource congestion.

Step 2 Start transport link flux monitoring on the eGBTS LMT. Choose **Monitor > Realtime Performance Monitoring > Transport Link Flux Monitoring**.

Step 3 Initiate a GSM or UMTS CS service if the traffic flux approaches the bandwidth available for the bearer network.

- Step 4** Terminate the CS service if the call is successfully set up and the voice is clear and constant.
- Step 5** Initiate a GSM PS service, connect a personal computer (PC) to the multimode base station, and use the traffic observation tool on the PC, for example, DU Meter to check whether the GSM PS service is successfully set up and the data rate is stable.
- If yes to both, this feature has been activated.
 - If no to either, this feature has not been activated.
- Step 6** Start IP or MAC tracing on the eGBTS LMT.
- For IP tracing: Choose **Trace > Common Services > IP Layer Protocol Trace**. In the displayed **IP Layer Protocol Trace** dialog box, specify **Local IP Address** and **Peer IP Address** of the packets to be traced.
- For MAC tracing: Choose **Trace > Common Services > MAC Trace**. In the displayed **MAC Trace** dialog box, specify **Local MAC Address** and **Peer MAC Address** of the packets to be traced.
- Step 7** Use the TrafficReview tool to check the TOS field in the layer 3 IP packet header or the VLAN Priority field in the layer 2 IP packet header. The first six bits in the TOS field indicate the DSCP value of a packet. If the calculated DSCP values or VLAN priorities are the same as the planned DSCP values or VLAN priorities, this feature has been activated.
- End

NodeB of a Multimode Base Station Providing a Co-transmission Port

- Step 1** Initiate a UMTS PS service and set the maximum data rate higher than the **CIR** value to simulate transmission resource congestion.
- Step 2** Start transport link flux monitoring on the NodeB LMT. Choose **Monitor > Realtime Performance Monitoring > Transport Link Flux Monitoring**.
- Step 3** Initiate a GSM or UMTS CS service if the traffic flux approaches the bandwidth available for the bearer network.
- Step 4** Terminate the CS service if the call is successfully set up and the voice is clear and constant.
- Step 5** Initiate a GSM PS service, connect a PC to the multimode base station, and use the traffic observation tool on the PC, for example, DU Meter to check whether the GSM PS service is successfully set up and the data rate is stable.
- If yes to both, this feature has been activated.
 - If no to either, this feature has not been activated.

NOTE

This step is performed only in a separate-MPT GU, GUL, or GULN multimode base station.

- Step 6** Start IP or MAC tracing on the NodeB LMT.
- For IP tracing: Choose **Trace > Common Services > IP Layer Protocol Trace**. In the displayed **IP Layer Protocol Trace** dialog box, specify **Local IP Address** and **Peer IP Address** of the packets to be traced.

For MAC tracing: Choose **Trace > Common Services > MAC Trace**. In the displayed **MAC Trace** dialog box, specify **Local MAC Address** and **Peer MAC Address** of the packets to be traced.

- Step 7** Use the TrafficReview tool to check the TOS field in the layer 3 IP packet header or the VLAN Priority field in the layer 2 IP packet header. The first six bits in the TOS field indicate the DSCP value of a packet. If the calculated DSCP values or VLAN priorities are the same as the planned DSCP values or VLAN priorities, this feature has been activated.

----End

eNodeB of a Multimode Base Station Providing a Co-transmission Port

- Step 1** Initiate an LTE PS service and set the maximum data rate higher than the **CIR** value to simulate transmission resource congestion.
- Step 2** Start transport link flux monitoring on the eNodeB LMT. Choose **Monitor > Realtime Performance Monitoring > Transport Link Flux Monitoring**.
- Step 3** Initiate a GSM or UMTS CS service if the traffic flux approaches the bandwidth available for the bearer network.
- Step 4** (Optional) Terminate the CS service if the call is successfully set up and the voice is clear and constant.
- Step 5** (Optional) Initiate a GSM PS service, connect a PC to the multimode base station, and use the traffic observation tool on the PC, for example, DU Meter to check whether the GSM PS service is successfully set up and the data rate is stable.
- If yes to both, this feature has been activated.
 - If no to either, this feature has not been activated.

NOTE

This step is performed only in a separate-MPT GL or GUL multimode base station.

- Step 6** Start IP or MAC tracing on the eNodeB LMT.

For IP tracing: Choose **Trace > Common Services > IP Layer Protocol Trace**. In the displayed **IP Layer Protocol Trace** dialog box, specify **Local IP Address** and **Peer IP Address** of the packets to be traced.

For MAC tracing: Choose **Trace > Common Services > MAC Trace**. In the displayed **MAC Trace** dialog box, specify **Local MAC Address** and **Peer MAC Address** of the packets to be traced.

- Step 7** Use the TrafficReview tool to check the TOS field in the layer 3 IP packet header or the VLAN Priority field in the layer 2 IP packet header. The first six bits in the TOS field indicate the DSCP value of a packet. If the calculated DSCP values or VLAN priorities are the same as the planned DSCP values or VLAN priorities, this feature has been activated.

----End

gNodeB of a Multimode Base Station Providing a Co-transmission Port

- Step 1** Initiate an NR PS service and set the maximum data rate higher than the **CIR** value to simulate transmission resource congestion.

- Step 2** Start transport link flux monitoring on the gNodeB LMT. Choose **Monitor > Realtime Performance Monitoring > Transport Link Flux Monitoring**.
- Step 3** Initiate a GSM or UMTS CS service if the traffic flux approaches the bandwidth available for the bearer network.
- Step 4** (Optional) Terminate the CS service if the call is successfully set up and the voice is clear and constant.
- Step 5** (Optional) Initiate a GSM PS service, connect a PC to the multimode base station, and use the traffic observation tool on the PC, for example, DU Meter to check whether the GSM PS service is successfully set up and the data rate is stable. If yes to both, this feature has been activated. If no to either, this feature has not been activated. This step is performed only in a separate-MPT GULN multimode base station.
- Step 6** Start IP or MAC tracing on the gNodeB LMT.
- For IP tracing: Choose **Trace > Common Services > IP Layer Protocol Trace**. In the displayed **IP Layer Protocol Trace** dialog box, specify **Local IP Address** and **Peer IP Address** of the packets to be traced.
- For MAC tracing: Choose **Trace > Common Services > MAC Trace**. In the displayed **MAC Trace** dialog box, specify **Local MAC Address** and **Peer MAC Address** of the packets to be traced.
- Step 7** Use the TrafficReview tool to check the TOS field in the layer 3 IP packet header or the VLAN Priority field in the layer 2 IP packet header. The first six bits in the TOS field indicate the DSCP value of a packet. If the calculated DSCP values or VLAN priorities are the same as the planned DSCP values or VLAN priorities, this feature has been activated.

----End

4.4.3.3 Limited Access Bandwidth for Each Operator in RAN Sharing Scenarios

- If you do not need to check whether the configured service priority has taken effect, perform the following steps to check whether the feature has been activated:
 - a. Run the **LST RSCGRP** (in the old model)/**LST IPRSCGRP** (in the new model) command to check whether the transmission resource group has been configured for the co-transmission port. If not, this activation observation method is not applicable.
 - b. Initiate a UMTS or LTE PS service for an operator and set the maximum data rate to a value greater than the **TXBW** value to simulate transmission resource congestion.
 - c. Query the value of the VS.RscGroup.TxFlowOverloadTime counter for the co-transmission port. If the value is greater than 0, this feature has been activated.
- If you need to check whether the configured service priority has taken effect, perform the following steps to check whether the feature has been activated:
 - a. Initiate a UMTS or LTE PS service for an operator and set the maximum data rate to a value greater than the **TXBW** value to simulate transmission resource congestion.

- b. Start transport link flux monitoring on the LMT.
 - If the NodeB side of a separate-MPT multimode base station provides a co-transmission port, start transport link flux monitoring on the NodeB LMT. Choose **Monitor > Realtime Performance Monitoring > Transport Link Flux Monitoring**.
 - If the eNodeB side of a separate-MPT multimode base station provides a co-transmission port, start transport link flux monitoring on the eNodeB LMT. Choose **Monitor > Realtime Performance Monitoring > Transport Link Flux Monitoring**.
- c. Initiate a UMTS CS service for operator A if the traffic flux approaches the bandwidth available for the bearer network. Terminate the CS service if the call is successfully set up and the voice is clear and constant.
- d. Perform the first three steps to verify services of other operators.
- e. Start IP or MAC tracing on the LMT.
 - Start IP or MAC tracing on the NodeB LMT.
For IP tracing: Choose **Trace > Common Services > IP Layer Protocol Trace**. In the displayed **IP Layer Protocol Trace** dialog box, specify **Local IP Address** and **Peer IP Address** of the packets to be traced.
For MAC tracing: Choose **Trace > Common Services > MAC Trace**. In the displayed **MAC Trace** dialog box, specify **Local MAC Address** and **Peer MAC Address** of the packets to be traced.
 - Start IP or MAC tracing on the eNodeB LMT.
For IP tracing: Choose **Trace > Common Services > IP Layer Protocol Trace**. In the displayed **IP Layer Protocol Trace** dialog box, specify **Local IP Address** and **Peer IP Address** of the packets to be traced.
For MAC tracing: Choose **Trace > Common Services > MAC Trace**. In the displayed **MAC Trace** dialog box, specify **Local MAC Address** and **Peer MAC Address** of the packets to be traced.
- f. Use the TrafficReview tool to check the TOS field in the layer 3 IP packet header or the VLAN Priority field in the layer 2 IP packet header. The first six bits in the TOS field indicate the DSCP value of a packet. If the calculated DSCP values or VLAN priorities are the same as the planned DSCP values or VLAN priorities, this feature has been activated.

4.4.4 Network Monitoring

None

5 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see the *Glossary*.

8 Reference Documents

1. *Transmission Resource Management*
2. *Common Transmission*
3. *SRAN Networking and Evolution Overview*
4. *HSDPA*